Denied Opportunity? An Analysis of Risk and Protective Factors Contributing to Incarceration Among Foster Care Youth

Abstract

This research paper aims to identify the top risk and protective factors contributing to incarceration of foster youth. This research used the AFCARS and NYTD outcomes file from the National Children's Bureau. Bivariate regression as well as correlation analysis was performed to select variables for the models. The models used were: baseline logistic regression, logistic regression with a class balanced dataset, logistic regression with a class balanced dataset removing states, support vector machine and random forest. The most accurate model was the random forest with an accuracy of 92.5%. The biggest risk factors across models were found to be: having 11 or more placements, child behavioral problems as reason for removal, substance abuse referral and current placement in an institution or group home. The biggest protective factors were found to be: being female, current placement in a pre-adoptive home or with a relative, and having less than 2 placements. While causality may not be inferred, there is a strong relationship between these factors and incarceration for which reason risk factors or lack of protective factors may be used to raise a red flag in the system if the system indicating extra attention may be needed.

I. Introduction

According to the Children's Bureau, there currently are over 407,000 children in the foster care system in the United States. While length of stay in the system varies one thing is certain, the uncertainty and lack of stability these kids face. Constant moving in placements prevents these individuals from succeeding in an academic environment as well as building meaningful relationships and bonds. The mix of both translates to foster care youth aging out of the system with very limited skills and support to navigate the transition (Osgood, 2010). In fact, a study localized in Chicago found that 14% of individuals face homelessness within the first six months of exiting the foster care system (Dworsky and Courtney, 2009). The lack of support and guidance many of these individuals receive during their time in the foster care system poses a great barrier to self-sufficiency as well as positive outcomes in the future.

Research has shown that foster care youth are more likely to face a host of negative outcomes such as incarceration. Once a kid has entered foster care, they are at a disadvantage in terms of the juvenile justice processing system as foster care status on its own can trigger a negative bias thus legal system involvement (Yang 2017.). Prior studies on legal involvement of young people in foster care

identified several risk factors. The risk factors primarily include demographic factors, such as being male and a person of color. The effects of this risk factor are large, as youth of color comprise the majority of the foster care system at a national level (Ryan, 2007). The National Youth Transition Database shows that over one third of foster youth surveyed had already been incarcerated by the age of 17. Defining incarceration as being confined in
a jail, prison, correctional facility, or juvenile or community detention facility in connection with allegedly committing a crime (misdemeanor or felony) (National Data Archive on Child Abuse and Neglect).

Being involved in the legal system at a young age has various adverse effects on various future opportunities. Having a criminal record greatly decreases the odds of items ranging from receiving college financial aid to receiving public housing benefits (Pager, 2003). A kid coming from a background where they were removed from their home already would likely have a disadvantage in these areas, but now simply because they have been removed from their home they face an increased negative bias in the juvenile justice system, which prompts legal involvements early in life, thus reducing opportunities further; not to mention how factors other than the legal system such as lack of stability contribute to this. For this reason, I would like to explore the risk and protective factors that increase the likelihood of incarceration once a kid has already entered the system. By having a better understanding of these factors it is possible to better shape policy or allocate resources to prevent this outcome.

II. Literature review

II.A. Foster Youth and Incarceration in California

The Child Adolescent Social Work Journal published a study titled Mitigating Risks of Incarceration Among Transition-Age Foster Youth: Considering Domains of Social Bonds (Park and Courtney, 2022). This study uses the CalYOUTH study survey to assess how different domains of social bonds may act as protective factors in predicting risk for later incarceration. This study focuses on 17 year olds transitioning to adulthood in the California foster care system. The underlying survey was collected in three waves, the first when individuals were 17, the second at age 19, and the third at age 21. The study examined the relationship between bonds assessed in wave one and the incarceration status of the same individuals in wave three.

The study classifies bonds into two main categories, interpersonal bonds and institutional bonds. Interpersonal bonds Institutional bonds refer to connections with family, caregivers and professionals. For each of these relationships, the questionnaire asked subjects to rate their closeness on a 1-5. Then these variables were made binary as by grouping 1-2 in 1-Close and 3-5 in 0-Not Close. In contrast, institutional bonds were captured via school enrollment and employment, the only options for these questions were 1-Yes and 0-No. Other predictive variables included in the study include demographic characteristics, maltreatment history prior to care, behavioral health problems, placement types, placement instability and time in extended foster care (wave two and three). These variables together with social bonds were used to build a logistic regression to estimate the risk of incarceration at the age of 17-21.

After bivariate analysis, only 3 bonds were found to be statistically significant, bonds to substitute caregiver, education and employment. These three variables together with other predictive variables were included in the logistic regression. Social bonds were not a significant predictor of incarceration later in life with the exception of bond with substitute caregivers which was negatively associated with incarceration odds with marginal statistical significance. For institutional bonds, school enrollment at 17 was associated with a 53% decrease in estimated odds of later incarceration and being employed at 12 was associated with a 57% decrease in estimated odds of later incarceration. Besides bonds, other relevant predictors include race, black individuals were 2.3 times as likely to be incarcerated than white individuals. In addition, males were found to be 4.3 times more likely to be incarcerated than females. Other factors that increased odds of later incarceration include symptoms consistent with drug/ alcohol abuse, age of entry in to care, history of sexual abuse and total number of placements.

II.B. Foster Youth and Homelessness Nationally

The Children Youth and Services Review published a paper titled "Risk and protective factors contributing to homelessness among foster care youth: An analysis of the National Youth in Transition Database" (Kelly, 2019). This paper uses the National Youth in Transition Database (NYTD) to analyze risk and protective factors that contribute to homelessness among transition aged foster care youth. While factors contributing to homelessness for this demographic are well researched, protective factors are not; specially at the national level. The NYTD includes many files relevant to foster care including the Adoption and Foster Care Analysis Reporting System (AFCARS) and their Outcomes survey. The AFCARS is collected yearly at the state level and reported while the Outcomes survey is collected every three years in three waves. Wave one includes individuals who are 17, wave two follows those individuals when they are 19, and wave three does the same at age 21. Wave one was used to establish the baseline and possible risk/ proactive factors while wave three was used to record the outcome; homelessness status.

After conducting logistic regression, the author finds that the strongest protection factor is connection to adults; having a connection to an adult decreases the odds of homelessness by 58.7%. Other predictive factors include education, if the individual has obtained a high school level education, their odds of becoming homeless decreased by 28.7%. Individuals enrolled in highschool had 22% lower odds of becoming homeless. Regarding risk factors, the strongest risk factor was having been incarcerated, this increased the odds of becoming homeless by 157.5%. Other risk factors include substance abuse problems which increased the odds of homelessness by 108.6%. In addition, having a runaway history increased odds of homelessness by 63.7%. Unexpectadly, this paper found that home removal due to sexual assault or inadequate housing actually decreased the risk of homelessness by 20.1% and 30% respectively rather than increasing it as expected by the researchers. This discrepancy is attributed to the small sample size of youth who had been removed due to sexual assault/ inadequate housing who had experienced homelessness.

III. Research Scope

I am hoping to explore which set of attributes are related to an increased risk of incarceration and which set of attributes are related to reduced risk of incarceration. By gaining an understanding of these factors, the foster care system could guide resource allocation. While resource allocation at the system level may be hard to justify as this is an exploratory paper thus causality will not be explored, resource allocation at the individual level could be guided. For instance, if an individual has X set of attributes which are a risk factor, they may need extra attention from their case worker. This process could be automated by the use of a risk prediction model which simply flags individuals at risk given the information they submit to the AFCARS on a yearly basis.

Based on the literature research, I am expecting to find that homelessness, lack of enrollment in school and being a male of color to be the biggest risk factors. Similarly, I am expecting for connection to adults, enrollment in school and placement with a relative to be the biggest protective factors. As far as model performance, I am expecting the random forest to outperform the logistic regression as this has been documented in risk prediction models for child welfare (Chouldechova, 2018).

IV. Data

IV. A. About the Data Set

This paper uses data from the National Children's Bureau. Firstly, we use the outcomes section of the National Youth Transition Database (NYTD). The NYTD outcomes collects demographic and "outcome" data from foster youth who have or are near to aging out of the system. The survey is conducted in three waves. The first wave tracks foster youth at 17. The second wave follows this same group at age 19 and the third wave surveys the same group at 21 ("NYTD Reporting

Systems."). States will survey youth regarding six outcomes: financial self-sufficiency, experience with homelessness, educational attainment, positive connections with adults, high-risk behavior, and access to health insurance. This database was born out of the John H. Chafee Foster Care Independence Program which allocates state governments flexible funding to provide programs that assist foster care youth in the transition to adulthood. In exchange, states must collect information on usage of various service usage as well as outcomes for foster youth. The data must be 90% error free which means that it must be free of missing information and invalid information ("About NYTD.").

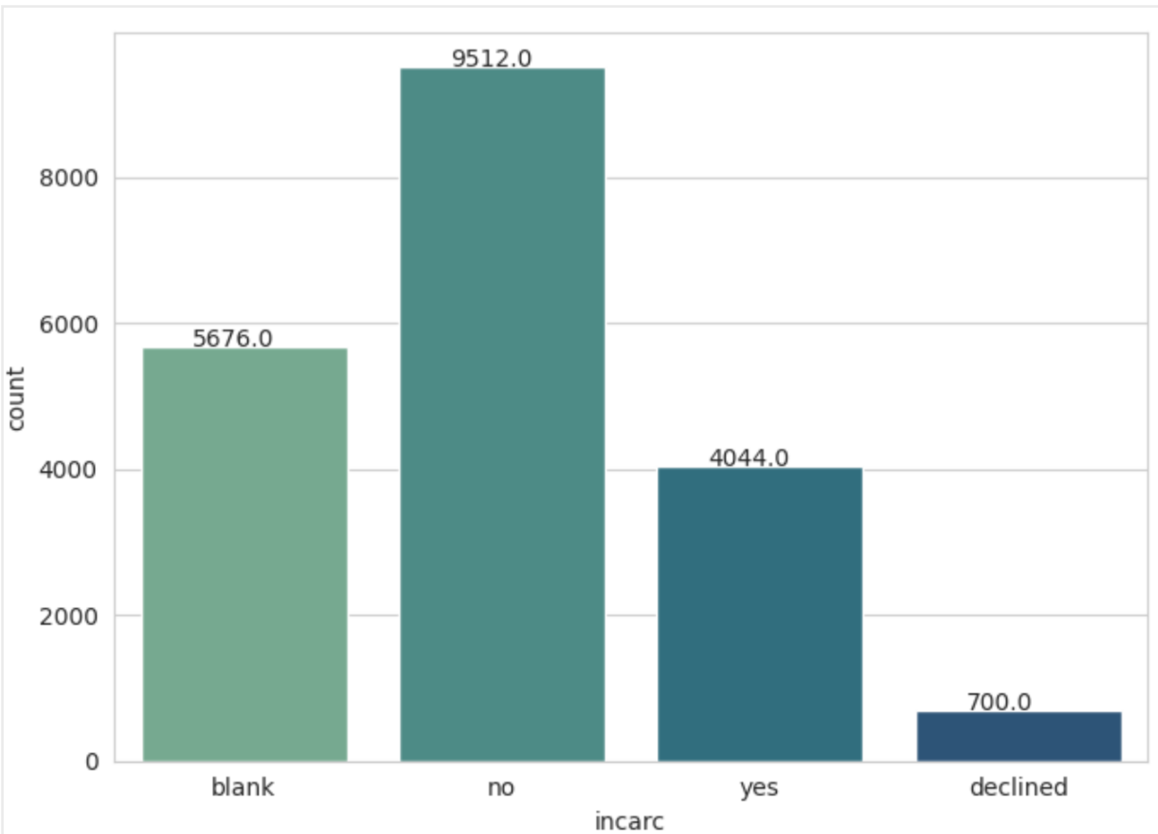
The second datasource from the National Children's Bureau is the Adoption and Foster Care Analysis and Reporting System (AFCARS). The Department of Health and Human Services (HHS), Administration for Children and Families (ACF), Children's Bureau (CB) is responsible for the implementation and management of AFCARS. All State and Tribal title IV-E agencies are required to report AFCARS case-level information every year ("About AFCARS."). This case level information includes demographics, reason for entrance into the system, number of placements, current placement status, medical disabilities, adoption history and more.

IV. B. Data Selection and Formatting

I decided to limit my study to individuals included in the most recent wave one of the NYTD outcomes file. This data was collected in 2017 and all individuals surveyed were 17 at the time. For consistency, I selected the AFCARS file from 2017.Since both databases contain an encrypted unique child ID. This common ID allows for both datasets to be merged. The data set now contains information for all individuals present in both datasets. After merging both datasets, I decided to limit my study to individuals included in wave one of the NYTD outcomes file because wave one has significantly more entries than two and three. Wave one has 22,609 entries; the subsequent waves have under 7,000 entries.

To assess potential missingness due to "declined" and "blank" answers on several columns, I calculated a "missingness score". For each individual, a missingness score corresponds to the number of blank or declined answers each individual answered per survey. For our current dataset, the mean number of questions with blank answers per person is 1.809 while the mean of questions with declined answers per person is 0.134. That being said, as incarceration is our dependent variable, it does not make much sense to keep blank and declined answers for incarceration in our data set as it does not provide much information regarding our research question. At this point, the distribution of responses in incarceration may be seen in Figure 1. After dropping entries that have blank or declined answers to incarceration our dataset reduces from 22,609 entries to 15,033 entries. That being said, the mean number of questions with blank answers per person reduces from 1.809 to 0.004 while the mean of questions with declined answers per person reduces from 0.134 to 0.059. This is a good sign as it indicates that we have reduced missingness in the rest of our dataset by dropping declined and blank responses in our dependent variable. This sharp decrease in missingness scores also indicates that it was the same individuals who declined and left incarceration blank as other variables. Since those who answered declined and blank for incarceration were dropped, there will be an analysis of the profile of these individuals in the results section.

Figure 1. Distribution of Incarceration Answers



The current values under consideration for the analysis are as follows:

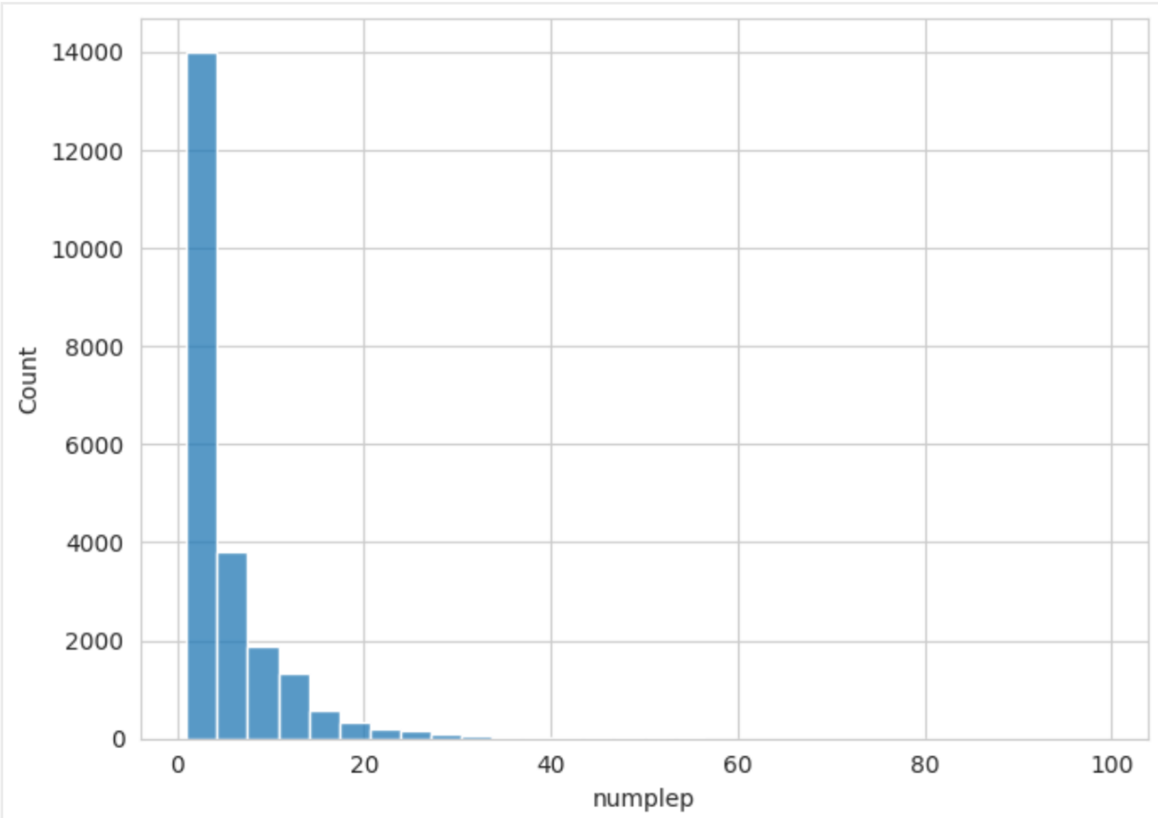
Figure 2: Variables Under Consideration
* need to be modified

Category	Name	Description	Range	Percentage (yes)
Demographics	St	Two-letter USPS code for state	AL-WY	n/a
	Sex	Gender is female	1: yes, 0: no	51.47%
	Amiakn	American Indian Or Alaskan Native	1: yes, 0: no	3.57%
	Asian	Asian	1: yes, 0: no	1.36%
	Blkafram	Black Or African American	1: yes, 0: no	33.05%
	Hawaiipi	Native Hawaiian Or Other Pacific Islander	1: yes, 0: no	0.74%
	White	White	1: yes, 0: no	62.79%
Outcomes	CurrEnroll	Current Enrollment And Attendance (academics)	1: yes, 0: no	93.00%
	CnctAdult	Connection To Adult	1: yes, 0: no	93.95%
	Homeless	Homelessness (in lifetime)	1: yes, 0: no	21.33%
	SubAbuse	Substance Abuse Referral (in lifetime)	1: yes, 0: no	24.01%
	Incarc **	Incarceration (in lifetime)	1: yes, 0: no	29.87%
Medical Conditions	ClinDis	Child Has Been Clinically Diagnosed with Disability	1: yes, 0: no	41.46%
	MR	Mental Retardation	1: yes, 0: no	2.99%
	VisHear	Visually or Hearing Impaired	1: yes, 0: no	3.96%
	PhyDis	Physically Disabled	1: yes, 0: no	0.82%
	EmotDist	Emotionally Disturbed	1: yes, 0: no	33.14%
	OtherMed	Other Medically Diagnosed Condition Requiring Special Care	1: yes, 0: no	13.92%
Placemet Information	TotalRem *	Total Number of Removals from Home	1-18	n/a
	NumPlep *	Placement Settings in Current FC Episode	1-93	n/a
	ManRem	Removal Manner	1: court ordered. 0: voluntary	94.49%
	CurPlSet	Current Placement Setting	1: pre-adoptive home , 2: foster home (relative) , 3: foster home (non-relative) , 4: group home , 5: institution , 6: supervised independent living , 7: runaway , 8: trial home visit	n/a
	PlaceOut	The Current Placement Setting is Outside the State	1: yes, 0: no	3.31%
Reason For Removal	PhyAbuse	Removal Reason-Physical Abuse	1: yes, 0: no	12.78%
	SexAbuse	Removal Reason-Sexual Abuse	1: yes, 0: no	8.42%
	Neglect	Removal Reason-Sexual Abuse	1: yes, 0: no	49.28
	AAParent	Removal Reason-Alcohol Abuse Parent	1: yes, 0: no	4.37%
	DAParent	Removal Reason-Drug Abuse Parent	1: yes, 0: no	13.80%
	AAChild	Removal Reason-Alcohol Abuse Child	1: yes, 0: no	1.24%
	DAChild	Removal Reason-Drug Abuse Child	1: yes, 0: no	4.48%
	ChilDis	Removal Reason-Child Disability	1: yes, 0: no	4.27%
	ChBehPrb	Removal Reason-Child Behavior Problem	1: yes, 0: no	32.27%
	PrtsDied	Removal Reason-Parent Death	1: yes, 0: no	1.44%
	PrtsJail	Removal Reason-Parent Incarceration	1: yes, 0: no	4.58%
	Abandmnt	Removal Reason-Abandonment	1: yes, 0: no	10.13%
	Relinqsh	Removal Reason-Relinquishment	1: yes, 0: no	2.59%
	Housing	Removal Reason-Inadequate Housing	1: yes, 0: no	6.84%

** target variable

As we plan to perform a logistic regression, we must one hot encode all of our independent variables. This process includes making all of our variables binary. To avoid collinearity, we must set a baseline variable. In this case, for each variable that has a “yes” and “no” answer, the baseline becomes the “no” answer. In this case, the one hot encoded version of some non-binary numeric values becomes quite extensive (i.e. 93 NumPlep variables, 18 variables for total removals). To regroup both of these variables, it is necessary to first understand their underlying distribution.

Figure 3. Distribution of Number of Placements



As seen in Fig 3, the number of placements follows an exponential decay distribution, approaching 0 around 25. For this reason, we decided to group NumPlemp in the following way for each entry:

If (NumPlep = 1) is true, then the new variable “numpla_1” is true thus equal to 1

If (NumPlep = 2) is true, then the new variable “numpla_2” is true thus equal to 1

If (NumPlep = 3) is true, then new variable “numpla_3” is true thus equal to 1

If (NumPlep = 4) is true, then the new variable “numpla_4” is true thus equal to 1

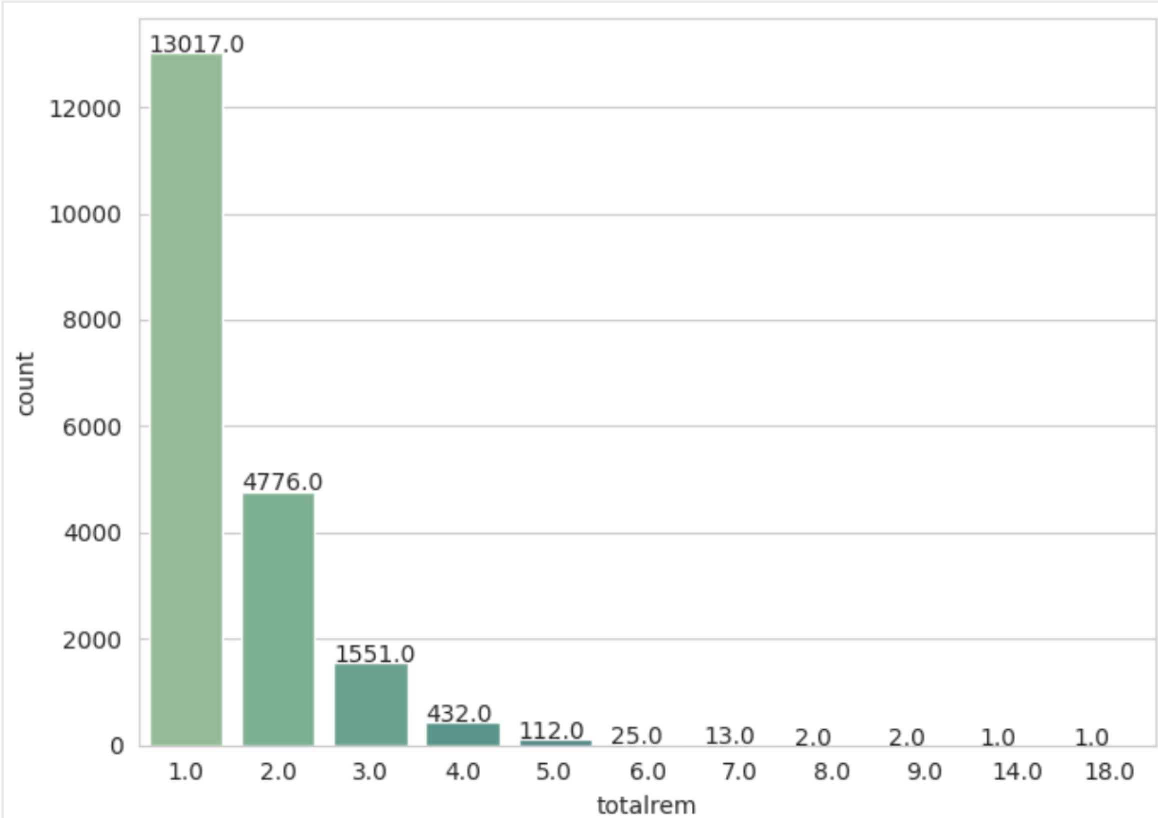
If (NumPlep = 5) is true, then the new variable “numpla_5” is true thus equal to 1

If (6≤ NumPlep≤ 10) is true, then the new variable “numpla_6_10” is true thus equal to 1

If (11≤ NumPlep) is true, then the new variable “numpla _11more” is true thus equal to 1

* If a “numpla_” variable does not equal 1, it equals 0 to indicate the condition is false

Figure 4. Distribution of Total Removals



Much like the number of placements, the distribution of total removals follows an exponential decay pattern. Since 1 is greater than the sum of all other placement columns, the total removals variable is regrouped as follows for each entry:

> If (TotalRem = 1) is true, then the new variable “removal_1” is true thus equal to 1
>
> If (2 ≤ TotalRem) is true, then the new variable “removal_2more” is true thus equal to 1
>
> * If a “removal_” variable does not equal 1, it equals 0 to indicate the condition is false

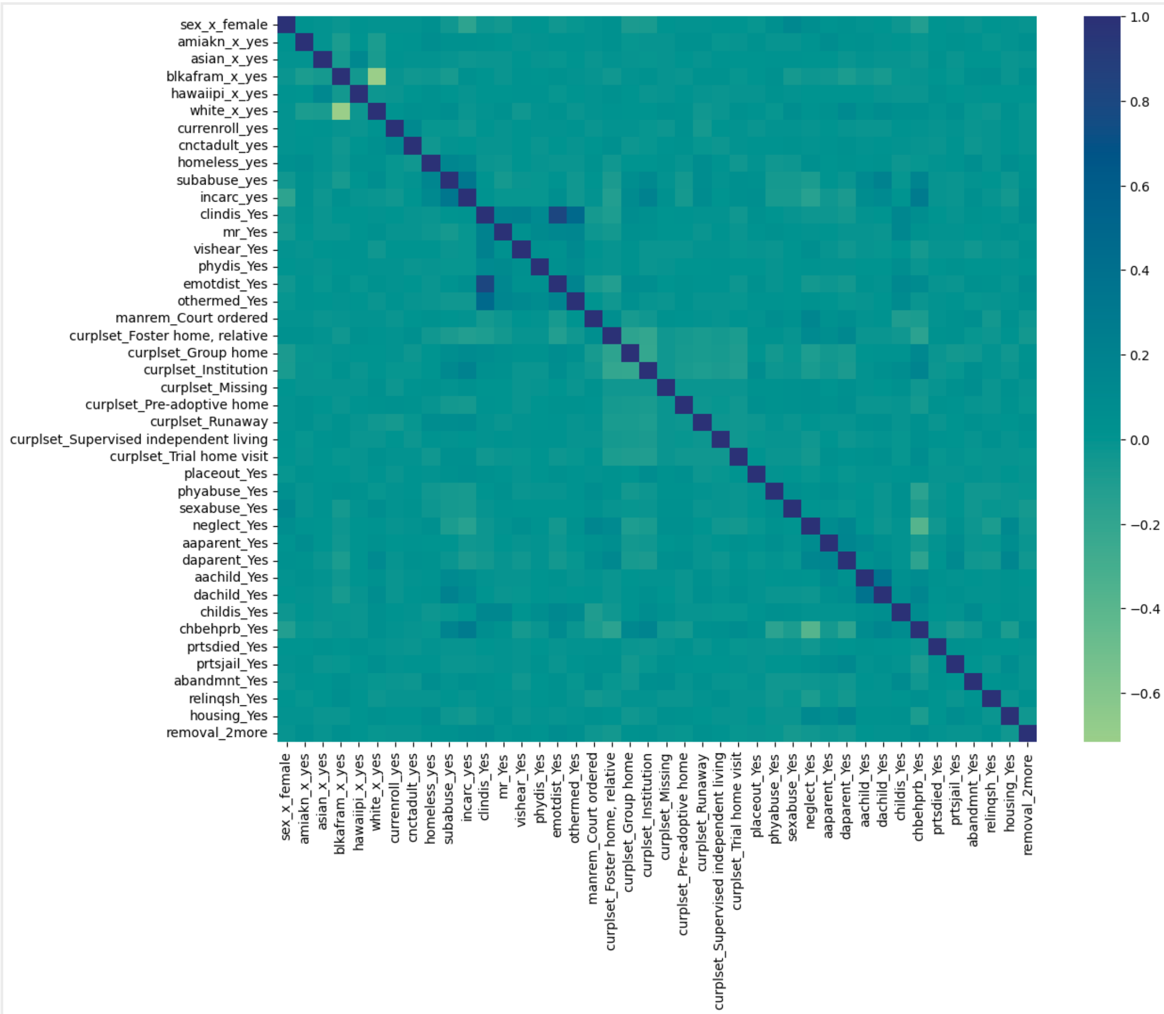
Since TotalRem was condensed into two variables, these will have perfect collinearity, therefore “removal_1” will be dropped and become the baseline.

IV. C. Selecting Variables for the Logistic Regression Model

After conducting bivariate logit regression for all variables (excluding state) on incarceration, we note that each of the listed variables has a $p < 0.1$ at significance level of $\alpha = 5\%$ with the exception of amiakn_yes [$p = 0.213$ at $\alpha = 5\%$], curplset_missing [$p = 0.149$ at $\alpha = 5\%$], and childis_yes [$p = 0.333$ at $\alpha = 5\%$]. Childis_yes will be dropped for the final regression but amiakn and curplset_missing will not be dropped as they are one of many “dummy variables'' for race and current placement respectively. The full table of results may be found in Appendix A.

After conducting a correlation analysis, we get the plot shown in figure 5.

Figure 5. Correlation Matrix for All Remaining Variables



As there are many variables not displayed in this heat map, we iterated through the matrix to find the most correlated pairs, the results are as follows:

- (clindis_yes, emotdist_yes): 0.809
- (blkafram_x_yes, white_x_yes): -0.7163
- (clindis_yes, othermed_yes): 0.471

While white_x_yes and blkafram_x_yes are highly correlated, since neither corresponds to a perfect or near perfect complement of the other, we will still include both variables in the logistic regression. This high correlation makes sense as they represent the two largest groups racially represented in our dataset. On the other hand, clindis_yes is correlated with both emotdist_yes and othermed_yes. Besides being correlated with two variables, it was also not a statistically significant regressor in the bivariate logistic regression, both of these confirm that clindis_yes should not be included in the logistic regression.

Similarly, we may take a deeper look at the variables that are most highly correlated with a positive response to incarceration (incarc_yes).

Figure 6. Correlation Between Independent Variables and Incarceration

Var	Corr	Var	Corr
subabuse_yes	0.317	abandmnt_Yes	0.038
chbehprb_Yes	0.279	othermed_Yes	-0.036
curplset_Institution	0.198	curplset_Trial home visit	0.035
sex_x_female	-0.177	blkafram_x_yes	0.033
neglect_Yes	-0.152	aaparent_Yes	-0.032
curplset_Foster home, relative	-0.118	asian_x_yes	-0.031
homeless_yes	0.118	amiakn_x_yes	0.026
dachild_Yes	0.106	vishear_Yes	-0.024
curplset_Group home	0.096	phydis_Yes	-0.023
curplset_Runaway	0.092	curplset_Missing	0.022
daparent_Yes	-0.078	mr_Yes	-0.021
sexabuse_Yes	-0.070	white_x_yes	-0.016
aachild_Yes	0.070	prtsjail_Yes	-0.016
phyabuse_Yes	-0.069	relinqsh_Yes	-0.011
removal_2more	0.060	clindis_Yes	0.010
curplset_Pre-adoptive home	-0.057	prtsdied_Yes	-0.009
placeout_Yes	0.052	hawaiipi_x_yes	-0.007
emotdist_Yes	0.051	manrem_Court ordered	0.005
housing_Yes	-0.047	childis_Yes	-0.003
currenroll_yes	-0.045	curplset_Supervised independent living	-0.002
abandmnt_Yes	0.038	cnctadult_yes	0.000

We may see that the most correlated risk factors are substance abuse (subabuse_yes), child behavior as reason for removal (chbehprb_Yes), and institution as current placement (curplset_Institution). The most correlated protective factors are female (sex_x_female), neglect as reason for removal (neglect_Yes) and a relative's home as current placement (curplset_Foster home, relative). It is important to remember that correlation does not imply causation, all we can interpret from this table is that in the case of risk factors, individuals who answer yes to these tend to also answer yes to having been incarcerated; we can make no assumptions about causation in either direction.

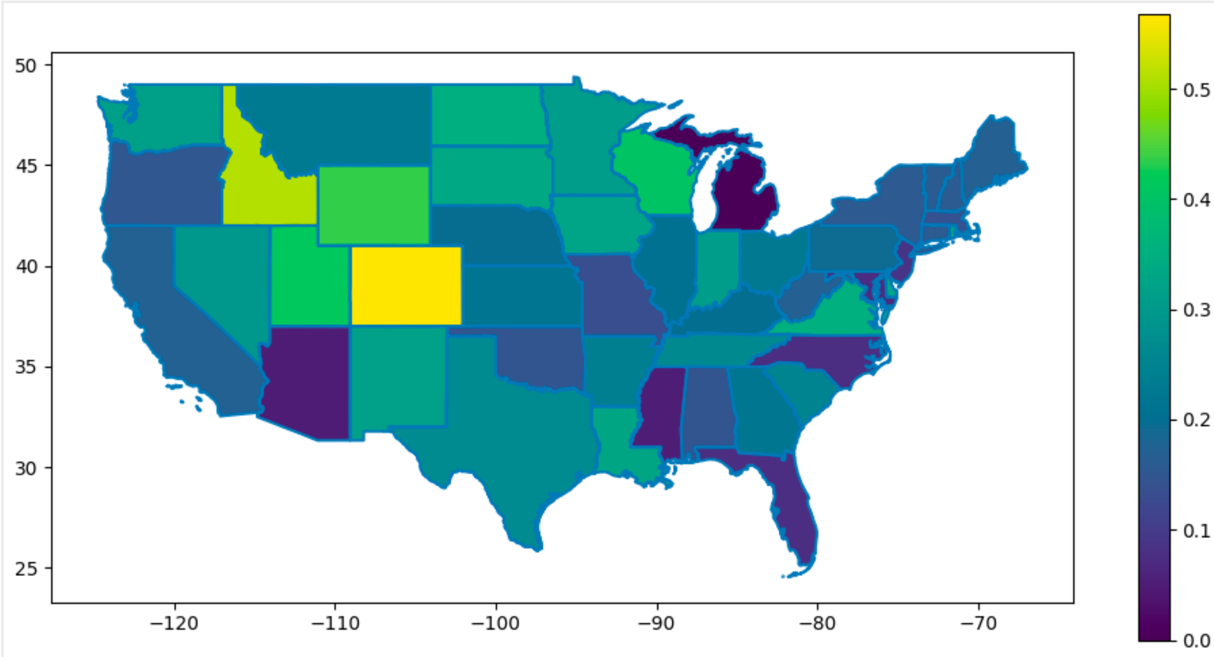
In terms of state variables, after computing a correlation matrix for all variables including states, we find that states are greatly uncorrelated with other variables. The only correlations involving states that have a correlation with an absolute value over 0.25 are as follows:

- (st_x_HI, hawaiipi_x_yes): 0.348
- (st_x_TX, childis_Yes): 0.265

The state of Hawaii and the race of hawaiian or pacific islander as expected to be correlated given that Hawaii has the highest proportion of individuals who identify as hawaiian or pacific islander. The Texas and child disability correlation coefficient may provide some insight into different removal thresholds or characteristics per state. As far as states and incarceration, only one state had a correlation coefficient with an absolute value above 0.1 with "incarc_yes". The highest correlation coefficients between a state and "incarc_yes" are as follows:

- Colorado: 0.107
- Ohio: 0.072
- Tennessee: 0.067
- Maryland: -0.065
- Utah: 0.055

Figure 7. Proportion of sample of each states answering "yes" to incarceration



After analyzing the incarceration response distribution per state, we may see that Colorado has the highest percentage of people who answered yes (56.79%). Ohio places 23rd at 22.43%, Tennessee 19th at 26.53%, Maryland 47th at 7.12% and Utah 3rd at (41.52%). It is expected to see Colorado at the top and Maryland at the bottom given the correlation values; this indicates that Colorado may appear as a risk factor while Maryland may appear as a protective factor in our analysis. This would not imply a causal relationship. It is also important to mention that the states that had a higher proportion of individuals answering "yes" to incarceration also had lower rates of "blank" and "declined" responses. As we have dropped "blank" and "declined" responses, we have added an extra bias to our state variable for which reason they may not be the most reliable. Missingness with respect to incarceration will be discussed in the results section.

V. Methodology

<u>V.A. Logistic Regression</u>

V.A.I. Baseline Logistic Regression

First, we run a logistic regression with all the variables in figure 6 with the exception of childis_yes. Our regression has the form:

log(incarceration) = +(state) +(female)+(race)+(outcomes) (medical conditions) + (removal hist) +(placement)+ (reason for removal)

Where,

- is a (51x1) vector with the coefficients for all the state dummy variables
- is the the coefficient for female
- is a (5x1) vector with the coefficients for all the race dummy variables (amiakn_yes, asian_yes, blkafram_yes, hawaiipi_yes and white_yes)
- is a (4x1) vector with the coefficients for all the outcome dummy variables (currenroll_yes, cnctadult_yes, homeless_yes, subabuse_yes)
- is a (5x1) vector with the coefficients for all the medical condition dummy variables (mr_yes, vishear_yes, phydis_yes, emotdist_yes, othermed_yes)
- is a (10x1) vector with all the coefficients for the removal history dummy variables (numpla_1, numpla_2, numpla_3, numpla_4, numpla_5, numpla_6_10, numpla_11more, removal_2more, manrem_courtordered, placeout_yes)
- is a (8x1) vector the coefficients for current placement setting dummy variables (curplset_pre-adoptive home, curplset_foster home (relative), curplset_fosterhome (non-relative), curplset_group home, curplset_institution, curplset_supervised independent living, curplset_runaway, curplset_trial visit)
- is a (14x1) vector of all the coefficients for the removal reason dummy variables (phyabuse_yes, sexabuse_yes, neglect_yes, aaparent_yes, daparent_yes, aachild_yes, dachild_yes, childis_yes, chbehprb_yes, prtsdied_yes, prtsjail_yes, abandoment_yes, relinquish_yes, housing_yes)

Besides obtaining the coefficients from our standard logistic regression, we will also run a K-fold validation to further assess the accuracy of our estimates. In this case, we will set our K-value to 5. This means that our data will be split into a training and a testing set 5 times, each time, the model will be fitted (we will obtain coefficients), and each time, the model will be tested with those coefficients; thus yielding us 5 different accuracy scores allowing us to calculate the average accuracy score and reduce the likelihood of a false strong correlation or R^2 due to overfitting.

V.A.ii. Logistic Regression with SMOTE-ENN data set

As we have a large class imbalance (e.g. 62.7% of the sample is white), it might cause for our regression to overfit to the majority classes, or for one small correlation due to small sample size to be blown out of proportion. For instance, only 0.74% of our sample identifies as hawaiian or pacific islander, this means that if 50 of the 121 individuals who are hawaiian happen to have had a history of incarceration (not representative of the overall population, just this sample), the model will overestimate the relationship between this race and incarceration when the relationship might not be there in the real world, it could just be a result of bias in the sample. This is just an example of how a relationship between an independent variable and the target may be overestimated due to small sample size; this could happen for any variable that is underrepresented in the sample. For this reason, we may choose to resample.

Synthetic Minority Oversampling Technique (SMOTE) is a statistical technique for increasing the number of cases in your dataset in a balanced way (Microsoft.). In general, SMOTE does not affect the instances of majority cases, only minorities. The algorithm generates new instances of minority samples by combining features of the minority feature and its nearest neighbors. We used a modification of SMOTE called SMOTE-ENN (Smote Edited Nearest Neighbors), this modification allows for the oversampling in the regular SMOTE as well as an additional undersampling of the majority class to minimize the number of new instances generated.

The logistic regression will be the same as in the previous section (V.A.I. Baseline Logistic Regression). The goal of this is to see how coefficients and accuracy change with a more balanced dataset. Though it is important to mention that by generating synthetic values, no matter how well interpolated, we may be introducing error. That being said, a biased data set will also produce error in a logistic regression.

V.A.iii. Smote Logistic Regression Removing States

As discussed in the data section, levels of “blank” and “declined” responses vary significantly by state. Some states such as New Mexico and Vermont have 0% of individuals answering “blank” or “declined” to the incarceration question. In contrast, other states such as Michigan have 100% of individuals answering “blank” or “declined” to the incarceration question. As we dropped individuals with who answered “blank” or “declined” to our target variable (given that it wasn’t helping answer our research question), we introduced a fair amount of potential bias, penalizing states with low “blank” or “declined” answer rates. By this logic, states with high missingness could be artificially placed as strong protective factors, or misrepresented as stronger than they are. As state seems to be a noise variable, we wanted to test how our results would change if the question was not involved in the regression. This logistic regression without states will follow the same function as the baseline and smote logistic regressions. The regression will be ran on both the baseline and the smote modified dataset to better assess the nature of the potential error introduced by the noise in the state variable.

V.B. Support Vector Machine

Next, we will use a Linear Support Vector Machine (Linear SVM) to assess the relationship between various risk/ protective factors and incarceration. One of the main advantages of SVMs is that they perform well when there's many input variables as is the case in this question. In addition, SVMs are very effective in high dimensional spaces which is a great advantage for this case as there are many variables. An SVM constructs a set of hyper-planes in high or infinite dimensional space, the goal of an SVM is to find a hyperplane that can separate data accurately (Fan, 2018). In this case, the data would ideally be separated by a hyperplane such that incarc_yes=1 for all points on one side of the hyperplane and incarc_yes=0 for all points on the other side of the hyperplane.

As Linear SVMs select a hyperplane, we may then get coefficients that represent the importance of certain variables relative to the others, as a greater magnitude can be interpreted as affecting the hyperplane positioning more. The relative importance of various risk and protective factors will be compared to that implied by the logistic regression. For this model, we will also use K-fold, selecting 5 different training and testing sets thus concluding 5 different hyperplanes, then taking the average. Performing K-fold reduces the risk of an artificially high accuracy due to overfitting.

Given the great class imbalance in the original data set, the likelihood of error due to bias or overfitting is likely larger than the error introduced by resampling, Therefore, this model will be explored with the SMOTE data set. This will hold for other machine learning methods in this project such as the Random Forest Classifier.

V.C. Random Forest Classifier

A Random Forest Classifier uses various decision trees to reach an output. In a decision tree, the leaf node of the path chosen will dictate the prediction. The training data going into each decision tree is random, and results may vary (Breiman). A random forest takes the majority vote of many decision trees to make a prediction; for this reason, it is best to have many trees feeding into the decision of the random forest. In a Random Forest Classifier out output coefficients are unitless thus difficult to compare to the coefficients from the logistic regression. That being said, much like in the case of SVMs, we can still perform feature importance analysis to assess the importance of features (independent variables, predictors) relative to others.

VI. Results and Discussion

VI.A. Logistic Regression

V.A.I. Baseline Logistic Regression

After conducting our baseline logistic regression, we see that the strongest risk factors for youth incarceration include substance abuse, child behavioral problems as the reason for removal, current placement setting in an institution and having had 11 or more placements. As seen in Figure 8, an individual who has had a substance abuse referral before 17 is over 2.5 times more likely to have been incarcerated by 17 than someone who has not received the same referral. While this does not imply causality, it may be useful information to prevent incarceration in the case an individual has received a referral but has not been incarcerated. Information about specific dates for these events would be needed in the form to possibly assess causality. Similarly, an individual who entered the system due to child behavioral problems is more than twice as likely to be incarcerated by 17 than an individual who entered the system for any other reason. An individual who has had 11 or more placements by 17 is 1.7 times more likely to have been incarcerated by 17. In this case, the question becomes, do these individuals get moved because of behaviors that lead to incarceration or does the lack of consistency due to moving lead to more adverse behavior which leads to incarceration? Both could be true, a case study research could provide more understanding into the nuance of the interaction of these factors.

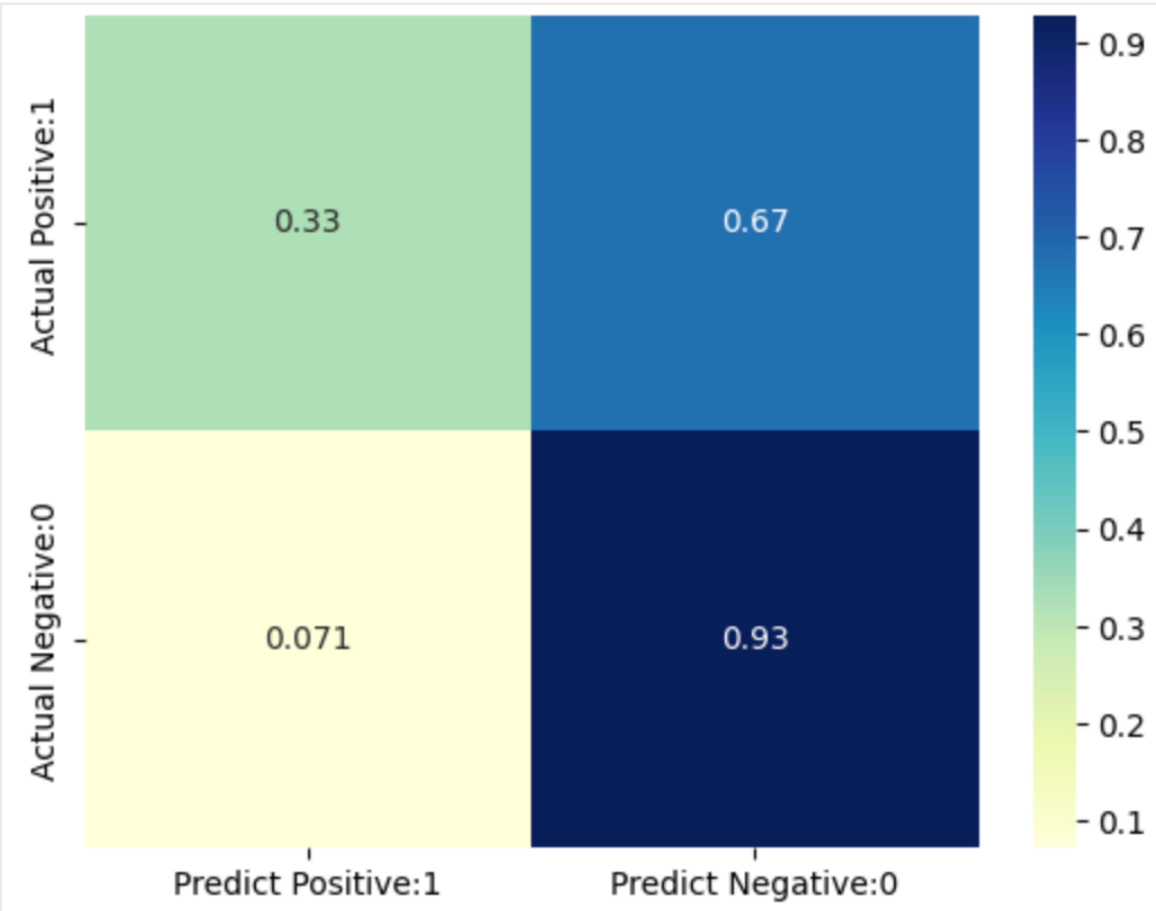
Figure 8. Top 10 protecting and top 10 risk factors for the baseline logistic regression

Risk Factors		**Protective Factors**	
var	**e^coef**	**var**	**e^coef**
subabuse_yes	2.5750	sexabuse_Yes	0.8530
chbehprb_Yes	2.0365	st_x_NJ	0.8503
curplset_Institution	1.7669	phyabuse_Yes	0.8488
numpla_11more	1.6032	st_x_PA	0.8243
homeless_yes	1.4606	st_x_MA	0.7979
curplset_Group home	1.4102	numpla_2	0.7959
curplset_Runaway	1.4080	st_x_MO	0.7869
st_x_OH	1.2951	st_x_MD	0.7817
st_x_TX	1.2878	neglect_Yes	0.7479
st_x_CO	1.2528	sex_x_female	0.5673

Surprisingly, the strongest protective factor was sexual abuse as reason for removal. Upon further investigation, this prominent placement may be attributed to a small sample size. Park and Courtney found this same relationship in their

2022 paper as mentioned in the literature review. The second largest protective factor, as youth residing in New Jersey is 15% less likely to be incarcerated. As seen in Appendix C, New Jersey is one of the states with the lowest “declined” and “blank” response rates, this makes the prominence of the state as a protective factor less likely to be due to bias in response rates; though it could be a relatively small sample size. We find that women are roughly 44% less likely to be incarcerated by the age of 17 than their male counterparts. This could be due to differences in behavior or gender bias in the legal system; regardless, it remains that males are at higher risk for incarceration.

Figure 9. Baseline Logistic Regression Accuracy Distribution



The baseline logistic regression model had an accuracy of 63.7%, this means that our model properly classified positive (1: incarceration) and negative (0: no incarceration) cases 63.7% of the time. That being said, the distribution of this accuracy is far from even. As seen in Figure 9, the baseline logistic regression model only classifies positives properly 33% of the time while it classifies negatives properly 93% of the time. This difference may be due to overfitting to the negative case as it represents the majority of our sample. A model with this bias would lead to a large amount of under-reporting of foster youth in risk, as 67% of individuals that had been incarcerated were predicted to be non-incarcerated.

V.A.ii. Logistic Regression with SMOTE-ENN data set

After conducting logistic regression on our smote-enn dataset, we may see that the top 4 risk factors remain the same, that being said, the magnitude of the coefficients increased. An individual who has received a substance abuse referral went from being 2.5 times more likely to have been incarcerated by 17 to being 3.2 times more likely to be incarcerated by the age of 17. Similarly, an individual who was removed due to behavioral problems went from being 2 times as likely to be incarcerated by 17 to being 2.7 times more likely to be incarcerated by 17. This pattern implies that the baseline logistic regression may have underestimated the connection between the top risk factors and incarceration due to their relatively low representation in the original data set.

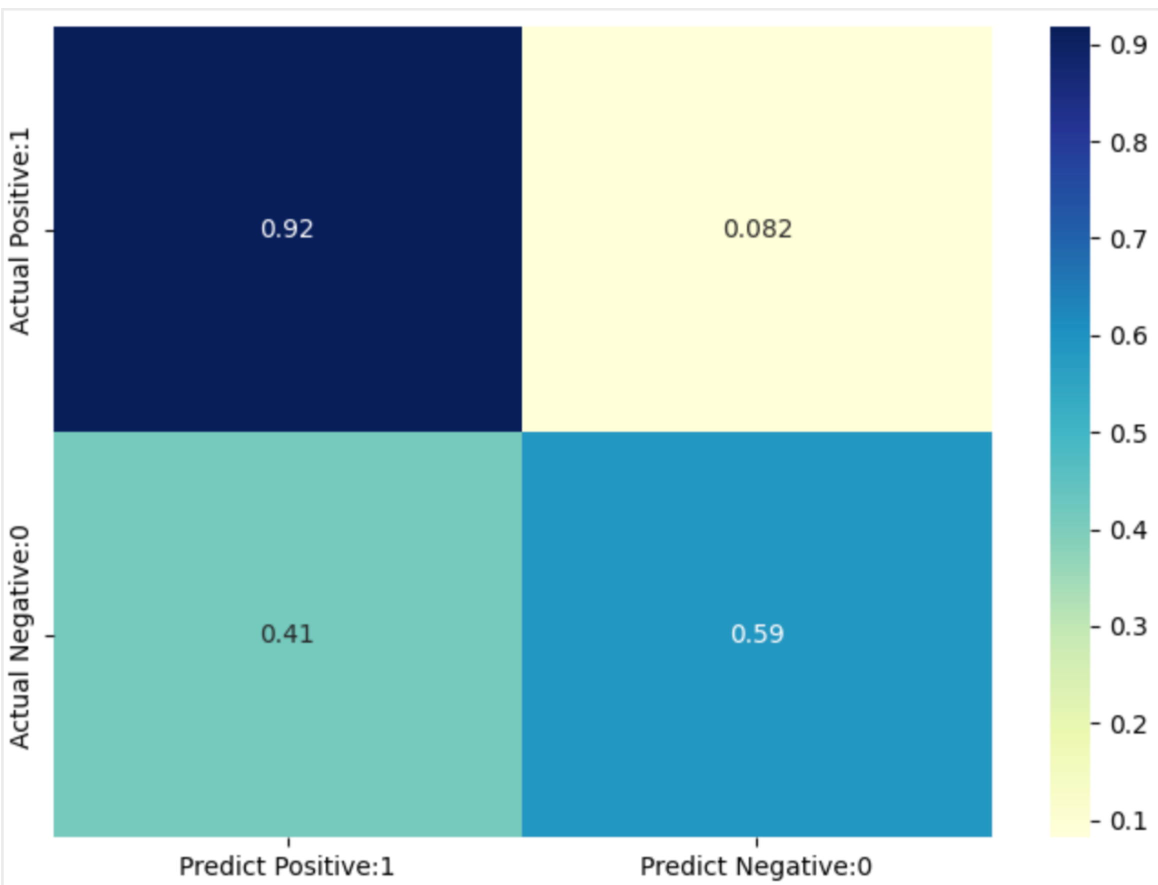
Figure 10. Top 10 protecting and top 10 risk factors for the smote-enn logistic regression

Risk Factors		**Protective Factors**	
var	**e^coef**	**var**	**e^coef**
subabuse_yes	3.2530	st_x_MA	0.7024
chbehprb_Yes	2.7450	othermed_Yes	0.6634
numpla_11more	2.4899	sexabuse_Yes	0.6456
curplset_Institution	1.8488	phyabuse_Yes	0.6429
curplset_Group home	1.4852	numpla_2	0.6326
homeless_yes	1.3937	daparent_Yes	0.6315
st_x_TX	1.3452	numpla_1	0.6256
numpla_6_10	1.2549	curplset_Foster home, relative	0.6009
emotdist_Yes	1.2510	neglect_Yes	0.5614
st_x_IN	1.1856	sex_x_female	0.3651

Regarding protective factors, we may see that now New Jersey is no longer in the top 10 list but now Massachusetts is in the top protective factor. As Massachusetts is the 8th state with the most missingness in responses, this seemingly strong relationship between no incarceration and the state is likely attributed to noise. In line with literature, a foster home place with a relative is one of the top protective factors (3rd) as individuals in this placement setting are 30%

less likely to be incarcerated by 17 than those in other placement settings. Having 1 placement makes the appearance of the top 10 now, this is in line with literature and suggests that its lack of appearance prior may be due to a relatively low prevalence in the data set. Individuals with 1 placement are 30% less likely to be incarcerated and those with 2 placements are 27% less likely to be incarcerated. Much like the discussion of 11 or more placements as a risk factor, the causality of the relationship between number of placements is hard to attribute as one a low number of placements could lead to stability thus lower odds of incarceration or individuals with less risky behavior may be moved less as their placement will have an easier time hosting them. The latter is an option as the policy in many placing agencies is to protect the home so they can keep fostering according to Franco Vega, the CEO of The Right Way Foundation. Again, being female remains the biggest protective factor. Using the smote-enn dataset, females are now 63% less likely to be incarcerated (as compared to the 44% suggested by the baseline regression).

Figure 11. Smote Logistic Regression Accuracy Distribution



By using the smote-enn data set, the accuracy of the logistic regression increased from 63.7% to 81.8%. We see an increase in the true positive rate from 33% to 92%, this means that 92% of the time the logistic regression saw an individual who would be incarcerated, it properly predicted the individual would be incarcerated. In line with this, we see a sharp decrease in the false negative rate from 67% to 8.2%. This rate represents the youth who was actually incarcerated but the model predicted they would not. If this model were used to allocate resources, a false negative would mean that an individual needed resources and didn't get them while a false positive indicates an individual did not need the extra resources but got them anyway. A balance between a low false positive rate and false negative rate is needed as an excessively small false negative rate at the cost of an excessively high false positive rate would be extremely wasteful (assuming budget constraints) and an excessively small false positive rate at the cost of an excessively high false negative rate while economically efficient, would leave lots of individuals who may have needed resources unattended.

V.A.iii. Smote Logistic Regression Removing States

Performing the logistic regression on the smote data set without states yields a similar list of risk factors as with states. The top risk factors are substance abuse, 11 or more placements, child behavioral issues as reason for removal and current placement in an institution. That being said, the magnitude of significance of these factors increases greatly. For instance, an individual with a substance abuse referral went from being 3.2 times more likely to be incarcerated by 17 to being 11.5 times more likely to to be incarcerated by the age of 17. It is possible that the state variable, while not interpretable or significant in terms of coefficients on its own, may have been accounting for state level variables that are not considered in other variables thus was aiding in understanding of the magnitude of the effects of other variables.

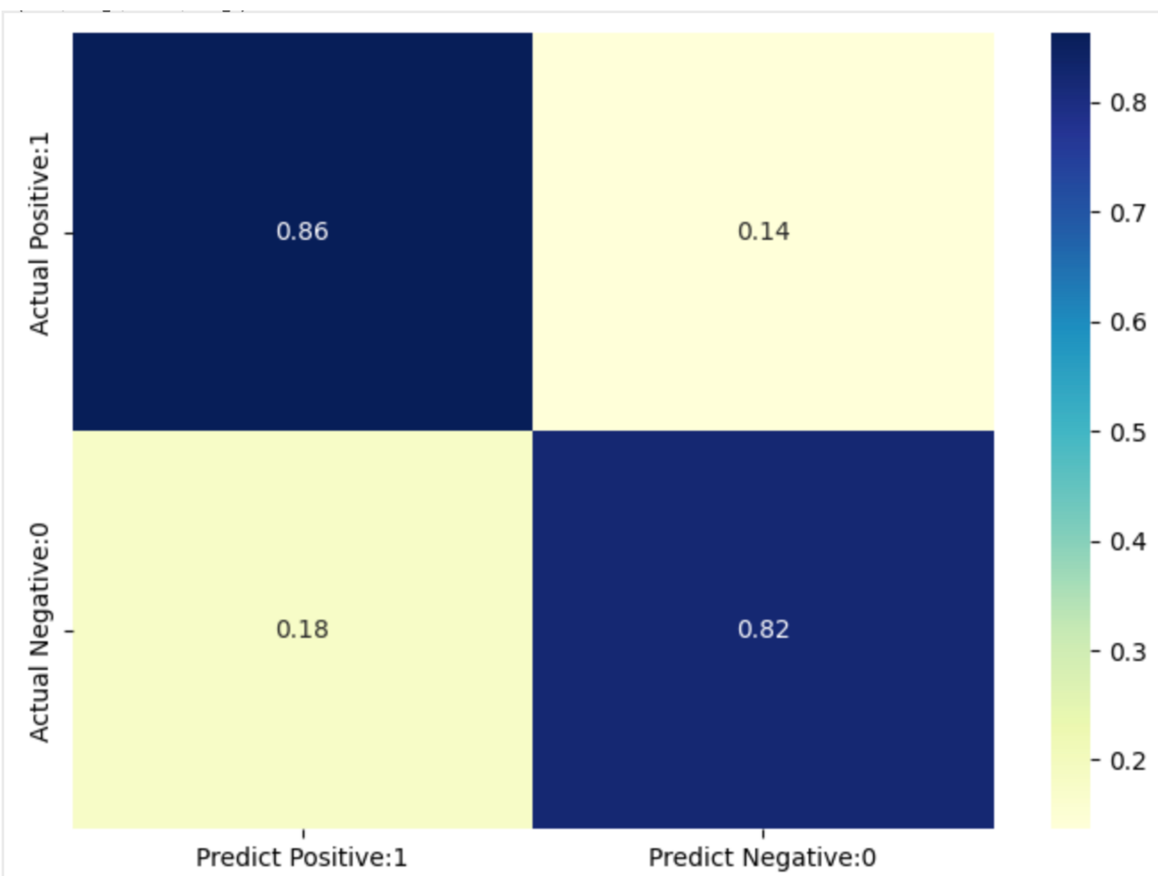
Figure 12. Top 10 protecting and top 10 risk factors for the smote-enn logistic regression (without states)

Risk Factors		**Protective Factors**	
var	**e^coef**	**var**	**e^coef**
subabuse_yes	11.4970	numpla_2	0.3482
numpla_11more	9.6733	curplset_Supervised independent living	0.3019
chbehprb_Yes	4.1781	prtsdied_Yes	0.2968
curplset_Institution	3.2759	mr_Yes	0.2797
curplset_Runaway	2.4372	hawaiipi_x_yes	0.2608

curplset_Group home	2.3167	relinqsh_Yes	0.2282
homeless_yes	2.2849	asian_x_yes	0.2155
aachild_Yes	1.8784	sex_x_female	0.1977
numpla_5	1.7348	curplset_Pre-adoptive home	0.1566
dachild_Yes	1.6734	phydis_Yes	0.1368

Regarding protective factors, being female remains a top protective factor. That being said, physical disability and placement in a pre-adoptive home enter the top 10 list. Much like in risk factors, the magnitude of these effects is greater as females went from being 63% less likely to be incarcerated (with states) to being 80% less likely to be incarcerated. In addition, for the first time we see race enter the top 10 factors with hawaiian or pacific islander and asian as protective factors. As race is not distributed evenly across states, this supports the idea that perhaps the state level variables were accounting for state level factors that were not present before thus helping us better understand the magnitude of the effects of other variables. The magnitude of the exact relationship between incarceration and other variables may vary across all the logistic regressions but the factors that are in the top 10 biggest risk and protective factors remain quite similar.

Figure 13. Smote Logistic Regression without States Accuracy Distribution



After removing states, the accuracy of the logistic regression increased from 81.8% to 85.7%. This means that in 85% of cases the logistic regression properly identified youth that was incarcerated by 17 and was not 85.7% of the time. We may see an increase in the balance between the false positive to false negative rate as with states, false positive was ~41% while false negative was ~8% and without states false positive is ~18% while false negative is ~14%. The increased accuracy implies that the state variable, while adding inside regarding the magnitude of the relationship between variables and incarceration, may have been adding noise to the overall classification. In addition, the increased balance in the false positive and false negative rate is preferred if this model were to be used for risk prediction as it better balances providing resources to those in need and budgetary constraints.

VII.B. Support Vector Machine (SVM)

Unlike a logistic regression, interpreting the coefficients of an SVM is not possible. That being said, these coefficients provide a great insight into the relative importance of each factor with respect to the others.

Figure 14. Top 10 protecting and top 10 risk factors for the SVM

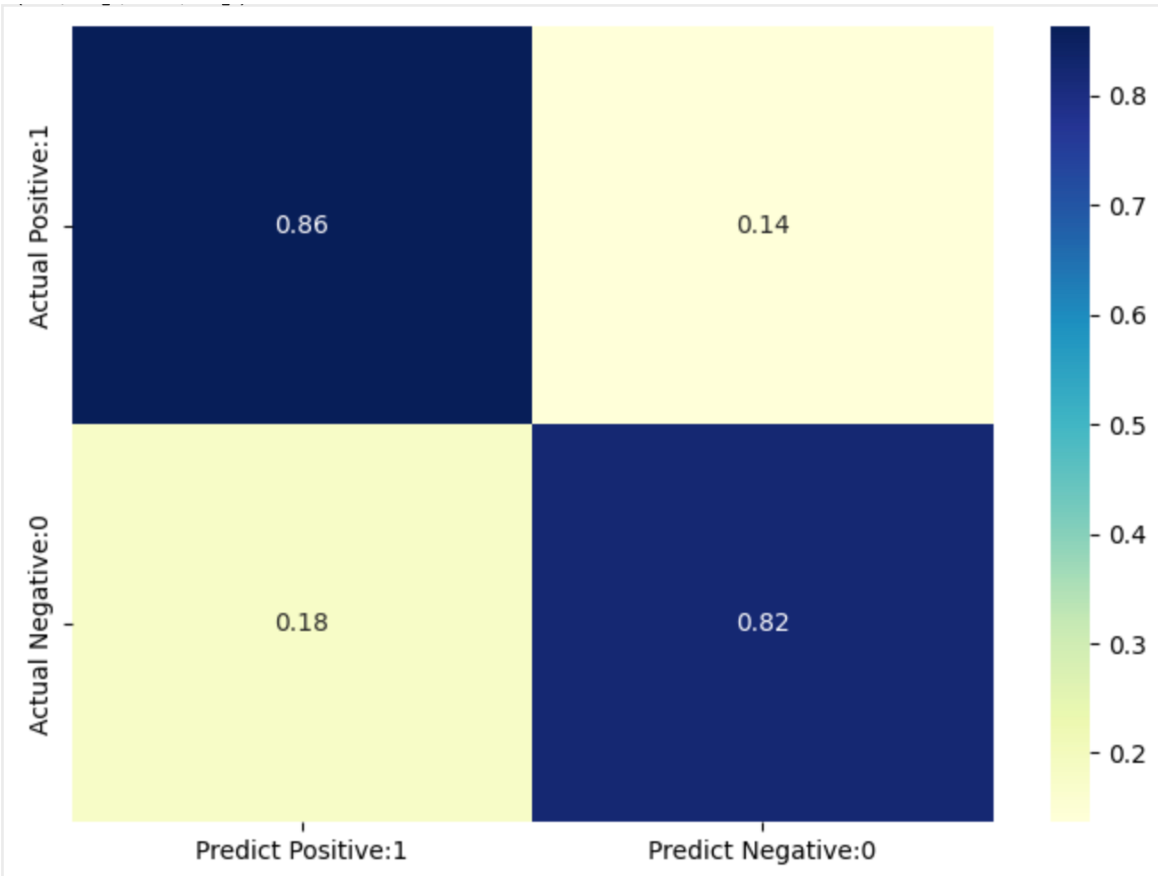
Risk Factors		**Protective Factors**	
var	**coef**	**var**	**coef**
numpla_11more	2.2560	phyabuse_Yes	0.7069
subabuse_yes	2.2165	curplset_Supervised independent living	0.6688
chbehprb_Yes	1.6260	prtsdied_Yes	0.6638
curplset_Institution	1.4886	hawaiipi_x_yes	0.6531
curplset_Runaway	1.3666	mr_Yes	0.6454
curplset_Group home	1.3425	relinqsh_Yes	0.6047
homeless_yes	1.3015	sex_x_female	0.5908
aachild_Yes	1.2553	asian_x_yes	0.5864
numpla_5	1.2542	curplset_Pre-adoptive home	0.5745
dachild_Yes	1.1852	phydis_Yes	0.4933

As seen in Figure 14, 11 or more placements, having a substance abuse referral, child behavioral problems as reason for removal and placement in an institution remain the biggest risk factor for incarceration by the age of 17. Similarly, gender and current placement in a pre-adoptive home remain the biggest protective factors. However, now physical disabilities, parents having relinquished rights and placement in supervised interdependent living enter the top 10 protective factor. As supervised independent living is only available after a certain age, this is very unlikely a causal relationship, unless the incarceration is taking place after eligibility for supervised independent living though that is timeline information we do not have.

Smote No State (Accuracy: 85.6%)

Figure 15. SVM Accuracy Distribution

The accuracy decreases from 85.7% to 85.6% when moving from logistic regression to an SVM with the same underlying dataset; though this difference is negligible. As seen in Figure 15, the accuracy for the SVM and the previously discussed logistic regression trained with the same data set seems to be extremely similar; thus the SVM holds the same use case merits of a balanced false positive and false negative class in addition to increased accuracy relative to prior models.



VII.C. Random Forest

In a random forest, the coefficient corresponds to the relative importance of the variable (whether it be in a positive or negative direction) thus we cannot distinguish between risk and protective factors. However, we may use the direction of the relationship between these variables in other models or correlation with incarceration to estimate whether they correspond to a risk or protective factor. We may see the most important factors contributing to incarceration in Figure 14.

Figure 14. Top 10 protecting and top 10 risk factors for the Random Forest

var	Relative importance
subabuse_yes	0.1309
chbehprb_Yes	0.0988
sex_x_female	0.0830
numpla_11more	0.0632
neglect_Yes	0.0550
curplset_Institution	0.0402
removal_2more	0.0306

curplset_Foster home, relative	0.0283
homeless_yes	0.0242
curplset_Group home	0.0239

In line with prior models, having a substance abuse referral by 17, child behavioral problems as the reason for removal, being female, and having more than 11 placements are the most prominent factors. We also see current placement settings in an institution and group home appear as in prior models, likely as a risk factor. Similarly, we see current placement with a relative appear as in logistic regression. In line with literature (but no other model) we see 2 or more removals appear as a relevant factor, likely as a risk factor.

Figure 17. Random Forest Accuracy Distribution

The random forest has the highest accuracy of all models explored with an accuracy: 92.5%. At first glance, this high accuracy makes it the most suitable model for use cases. While there is a disparity between the false positive rate (11%) and false negative rate (5.7%), even the “higher rate” (11%) is lower than the lowest false positive/ negative rate in any of the models with an “even” distribution of false positive and negative rate. While this model does predict allocating resources were not necessary more than it does predict withholding resources were necessary, it is not inefficient in a budget constraint point of view as it still has the lowest false positive rate (allocating resources were not necessary) than any other model explored with the exception of the baseline logistic regression which had a false positive rate of 7% but a false negative rate of 67%. Thus, if a model were to be used to assess the risk of incarceration by 17, a random forest would be the best. This result is consistent with the 2018 paper A case study of algorithm-assisted decision making in child maltreatment hotline screening decisions."

VII. Future Research

1. Introduction of time discontinuity
The Outcomes dataset is collected in 3 waves, at age 17, 19, and 21.I opted to perform all of my research with the first wave (age 17) as the dataset was reduced to under 7,000 responses. While choosing to keep my dependent and independent variable in the same time period conserved a sample size of over 20,000, it is hard to assess causality. Once an individual turns 19, the questions regarding the outcome variables such as incarceration change from asking if the individual has ever been incarcerated to, has the individual been incarcerated in the past 2 years. While my sample size would likely be smaller, collecting my target variable from the second wave would make causality more portable. For instance, say an individual's placement setting was institutional (one of the larger risk factors). Clearly, even if an individual, say, was placed in an institution (by 17) and then became incarcerated (between 17 and 19), it could still be correlation or they could both be driven by the same underlying variable. That being said, causality is more likely in that case than in the one currently being researched as under the current scope, an individual could have been incarcerated at 15 and then placed in an institutional setting at 17 as result. In this case having institutional placement as a "risk" factor would actually do very little to reduce risk of incarceration.

2. Conduct interviews of individuals determining placements
As placement setting as well as number of placements seem to have been a prevelent risk and protective factor, better understanding the logistics of placements, merits, etc, could be beneficial to better understanding the relationship between placements and incarceration, thus to what extent placement setting contributes to incarceration or if it's simply a result of it.

3. Case studies regarding the foster care system and incarceration
Since the data regarding timelines for incarceration and placements or removals is not available (and even if it was, assessing causality would still be tricky), this research could benefit from interviewing individuals who have been in the foster care system and incarcerated to better understand the drivers of incarceration. Questions could address reasons for removal from placements, resources used or lack thereof, specific instances that resulted in their incarceration amongst others.

This information would be crucial to complement the existing model and would likely result in fine tuning of variables included in the model and a better interpretation of the current results, an interpretation that would lead to a better use of the risk prediction model built. An interdisciplinary approach to these followup interviews would be greatly beneficial as not only a domain expert in child welfare as well as an expert in behavioral psychology could aid in the provision of even more nuanced insights regarding factors contributing to incarceration of individuals interviewed.

VIII. Conclusion

Risk factors: We find that overall, the factors that are most related with increased odds of incarceration by 17 across models are as follows:

- Having 11 or more placements by 17
- Child behavioral problems as reason for removal
- Having a substance abuse referral by 17
- Current placement in an institution
- Current placement in a group home

Protective factors: Similarly, we find that the factors that are most related with decreased odds of incarceration by 17 across models are as follows:

- Being female
- Current placement in a pre-adoptive home
- Current placement in a foster home with a relative
- Number of placements below 2

While more research is needed to better understand the relationship and potential causality between these factors and incarceration by 17, we can establish that there is definitely a relationship. This means that the risk predicitoon models could still be used to assess potential risk of incarceration, not on its own but as an aid to overloaded case workers. For instance, if an individual has been placed in an institution and has had more than 11 placements, but they haven't been incarcerated, it might be useful to raise a flag and see if there's a way to provide more resources.

In addition, the two time insensitive variables were gender and and child behavioral problems as reason for removal. In the case of gender, the reason for male being a risk factor is likely bias in policing/ justice system, that being said, that's not to say that perhaps males in the foster care system could get extra attention in this area as they are at greater risk. Maybe this attention takes the form of receiving knowledge about the justice system, maybe it means a sports league to keep them off the streets, but regardless of the method, it's raising a flag. Similarly for children removed from their homes due to behavioral issues, this is a status upon entry and it is likely this quality that is putting them at higher risk of incarceration. Perhaps seeing what placements children with this reason for entry usually have, and those who did not get arrested and attempting to mimic this, or perhaps extra support in school, an extra hour from their case worker to understand the situation could be beneficial.

As far as the time variant factors, we may not conclude whether the nature of their relationship with incarceration is causal. For instance, being placed in an institution could indicate increased risk for incarceration or it could indicate a lack of connection in the individuals life which leads to said placement and incarceration or it could even be a result of having a history of incarceration. That being said, regardless of the direction of the relationship, if an indvidual's placement setting is institutional and they haven't been incarcerated, it might be a sign that they need a little more support from the system relative to other individuals in order to prevent an adverse outcome.

We started with over 100 variables in our initial models, and these 9 risk and protective factors were the only that pertained their relevance across all models. More information may be needed to understand causality but regarding application of this research, having one or many of the 5 risk factors paired with lack of protective factors should be enough to riase a flag to case workers for increased risk of incarceration if they have not been incarcerated already.

X. Works Cited

"About NYTD." Administration for Children and Families, U.S. Department of Health & Human Services, 14 Jan. 2022, https://www.acf.hhs.gov/cb/fact-sheet/about-nytd.

"About AFCARS." Administration for Children and Families, U.S. Department of Health & Human Services, 14 Jan. 2022, https://www.acf.hhs.gov/cb/fact-sheet/about-afcars.

Breiman, Leo. "Random Forests." Machine Learning, vol. 45, no. 1, 2001, pp. 5-32. https://www.stat.berkeley.edu/~breiman/randomforest2001.pdf.

Chouldechova, Alexandra et al. "A case study of algorithm-assisted decision making in child maltreatment hotline screening decisions." FAT (2018).

Dworsky, Amy and Mark E. Courtney. "Homelessness and the Transition from Foster Care to Adulthood Among 19-Year-Old Former Foster Youth." Child Welfare, vol. 88, no. 4, 2009, pp. 23-56.

Fan, Shuzhan. "Understanding Mathematics Behind Support Vector Machines."

Shuzhanfan.github.io, 17 May 2018, https://shuzhanfan.github.io/2018/05/understanding-mathematics-behind-support-vector-machines/.

Kelly, Peggy. "Risk and protective factors contributing to homelessness among foster care youth: An analysis of the National Youth in Transition Database." Children and Youth Services Review, vol. 108, 2020, article 104589. ScienceDirect, doi:10.1016/j.childyouth.2019.104589.
Microsoft. "SMOTE - Azure Machine Learning | Microsoft Docs." Microsoft Docs, Microsoft, 2021, learn.microsoft.com/en-us/azure/machine-learning/component-reference/smote?view=azureml-api-2.

National Data Archive on Child Abuse and Neglect. "NYTD Outcomes Codebook." 2017. https://www.ndacan.acf.hhs.gov/datasets/pdfs_user_guides/nytd-outcomes-codebook.pdf.

"NYTD Reporting Systems." Administration for Children and Families, U.S. Department of Health & Human Services, 14 Jan. 2022, https://www.acf.hhs.gov/cb/research-data-technology/reporting-systems/nytd.

Osgood DW, Foster EM, Courtney ME. Vulnerable populations and the transition to adulthood. The Future of Children. 2010;20(1):209–229. doi: 10.1353/foc.0.0047.

Pager D. The Mark of a Criminal Record. American Journal of Sociology. 2003;108(5):937–975. doi: 10.1086/374403.

Park, K., Courtney, M.E. Mitigating Risks of Incarceration Among Transition-Age Foster Youth: Considering Domains of Social Bonds. Child Adolesc Soc Work J (2022). https://doi.org/10.1007/s10560-022-00891-z

Ryan JP, Herz D, Hernandez PM, Marshall JM. Maltreatment and delinquency: Investigating child welfare bias in juvenile justice processing. Children and Youth Services Review. 2007;29(8):1035–1050. doi: 10.1016/j.childyouth.2007.04.002. [CrossRef] [Google Scholar] [Ref list]

Yang J, McCuish EC, Corrado RR. Foster care beyond placement: Offending outcomes in emerging adulthood. Journal of Criminal Justice. 2017;53:46–54.

XI. Appendix

Appendix A. Results of Logit Bivariate Regression on Incarceration

	Var	P-Val
race	amiakn_x_yes	0.213
	asian_x_yes	0.000
	blkafram_x_yes	0.000
	hawaiipi_x_yes	0.000
	raceunkn_yes	0.000
	racedcln_yes	0.000
main demographics	sex_male	0.000
	currenroll_yes	0.000
	cnctadult_yes	0.004
	homeless_yes	0.000
	subabuse_yes	0.000
disabilities	clindis_yes	0.000
	mr_yes	0.000
	vishear_yes	0.000
	phydis_yes	0.000
	emotdist_yes	0.000
	othermed_yes	0.000
adoption	everadpt_unabletodetermine	0.045
	everadpt_yeschildhasbeenlegallya	0.000
total removals	totalrem_1	0.000
	totalrem_2more	0.000
	manrem_courtordered	0.000
current placement	curplset_fosterhomerelative	0.000
	curplset_grouphome	0.000
	curplset_institution	0.000
	curplset_missing	0.149
	curplset_runaway	0.000
	curplset_supervisedindependentli	0.006
	curplset_trialhomevisit	0.000
Reason for Removal	placeout_yes	0.000
	phyabuse_yes	0.000

	sexabuse_yes	0.000
	neglect_yes	0.000
	aaparent_yes	0.000
	daparent_yes	0.000
	aachild_yes	0.000
	dachild_yes	0.000
	childis_yes	0.333
	chbehprb_yes	0.000
	prtsdied_yes	0.000
	prtsjail_yes	0.000
	abandmnt_yes	0.000
	relinqsh_yes	0.000
	housing_yes	0.000
Number of Placements	numpla_2	0.000
	numpla_3	0.000
	numpla_4	0.003
	numpla_5	0.022
	numpla_6_10	0.000
	numpla_11more	0.000

Appendix B. Complete Correlation Matrix

	amiakn_x_yes	asian_x_yes	blkafram_x_yes	hawaiipi_x_yes	white_x_yes	currenroll_yes	cnctadult_yes	homeless_yes	subabuse_yes	incarc_yes	clindis_Yes
amiakn_x_yes	1.000000	0.002208	-0.091812	0.000146	-0.089851	-0.003350	0.014242	0.066919	0.034440	0.025538	0.017714
asian_x_yes	0.002208	1.000000	-0.053082	0.130916	-0.079799	0.011915	0.005633	0.003501	-0.008060	-0.031318	-0.013517
blkafram_x_yes	-0.091812	-0.053082	1.000000	-0.045737	-0.715569	-0.006063	-0.045410	-0.032006	-0.070263	0.032713	0.001076
hawaiipi_x_yes	0.000146	0.130916	-0.045737	1.000000	-0.026837	-0.012862	-0.000939	0.032869	-0.004830	-0.007051	-0.004768
white_x_yes	-0.089851	-0.079799	-0.715569	-0.026837	1.000000	0.009254	0.047777	0.031245	0.064863	-0.016299	0.004579
currenroll_yes	-0.003350	0.011915	-0.006063	-0.012862	0.009254	1.000000	0.123924	-0.025091	-0.023268	-0.044700	-0.002859
cnctadult_yes	0.014242	0.005633	-0.045410	-0.000939	0.047777	0.123924	1.000000	-0.040297	0.006067	-0.000307	-0.011392
homeless_yes	0.066919	0.003501	-0.032006	0.032869	0.031245	-0.025091	-0.040297	1.000000	0.160098	0.117607	-0.013276
subabuse_yes	0.034440	-0.008060	-0.070263	-0.004830	0.064863	-0.023268	0.006067	0.160098	1.000000	0.316696	-0.001512
incarc_yes	0.025538	-0.031318	0.032713	-0.007051	-0.016299	-0.044700	-0.000307	0.117607	0.316696	1.000000	0.009839
clindis_Yes	0.017714	-0.013517	0.001076	-0.004768	0.004579	-0.002859	-0.011392	-0.013276	-0.001512	0.009839	1.000000
mr_Yes	-0.000157	-0.017230	-0.014724	-0.001471	0.030246	-0.009910	-0.011120	-0.030467	-0.058543	-0.020815	0.204764
vishear_Yes	-0.011497	-0.000219	0.008942	0.014374	-0.043474	-0.001768	-0.002895	0.000090	-0.015082	-0.023641	0.210740
phydis_Yes	0.002261	-0.010697	-0.003106	0.009316	0.007824	0.004858	-0.013892	-0.011572	-0.020277	-0.022555	0.108363
emotdist_Yes	0.012210	-0.022726	0.016599	-0.019455	0.017505	0.002197	-0.003411	-0.013621	0.013442	0.050945	0.809009
othermed_Yes	0.010636	-0.002307	0.005112	-0.001003	-0.018503	0.010193	-0.011694	-0.015084	-0.033460	-0.036043	0.471106
manrem_Court ordered	0.007243	-0.017003	-0.008043	0.003816	0.006341	-0.014921	-0.000155	0.016930	-0.004042	0.004841	-0.059318
curplset_Foster home, relative	0.022361	0.005487	-0.038484	0.023977	0.014723	0.007382	0.042815	-0.038145	-0.072237	-0.117678	-0.103278
curplset_Group home	-0.022031	-0.006706	0.007500	-0.002105	0.001335	0.002198	-0.026354	-0.000133	0.065731	0.096248	0.075325
curplset_Institution	-0.006838	-0.019005	0.014959	-0.018299	0.001361	-0.024600	-0.021576	-0.000303	0.118802	0.198043	0.035868
curplset_Missing	-0.011240	-0.009550	-0.011755	-0.007022	0.026950	-0.029212	0.013754	0.017843	0.008137	0.022338	-0.005087
curplset_Pre-adoptive home	0.034325	-0.000027	-0.013680	0.011426	0.018063	0.026692	0.018600	-0.005976	-0.046908	-0.057496	0.012221
curplset_Runaway	0.026050	-0.005799	0.017026	-0.005327	-0.038353	-0.064574	-0.019562	0.068825	0.079877	0.091596	0.007696
curplset_Supervised independent living	-0.020322	0.008453	0.018376	-0.013355	-0.014575	-0.002316	-0.030850	0.008517	0.006664	-0.002051	0.007191
curplset_Trial home visit	-0.010803	-0.001756	-0.021327	-0.008235	0.026953	0.002213	0.006563	-0.037205	0.046409	0.035359	-0.036127
placeout_Yes	0.016531	-0.002394	-0.001011	-0.007256	0.008421	-0.017766	0.012569	0.008182	0.029309	0.051667	0.024116
phyabuse_Yes	-0.011402	0.011938	0.034747	0.018191	-0.020680	0.007462	0.010999	-0.033895	-0.056583	-0.069115	-0.001815
sexabuse_Yes	-0.008032	0.012050	-0.065413	0.004623	0.064469	0.006268	0.015628	-0.029819	-0.058326	-0.070196	-0.008222
neglect_Yes	0.009589	-0.002907	-0.025354	0.000468	0.010357	-0.010479	0.003874	0.047470	-0.087821	-0.151927	-0.028497
aaparent_Yes	0.071059	-0.002575	-0.041616	-0.003235	0.023210	-0.005074	0.010548	-0.002474	-0.005158	-0.032156	-0.026027
daparent_Yes	0.015500	-0.018580	-0.090842	-0.000707	0.108449	-0.008105	0.022181	0.020103	-0.016729	-0.078161	-0.048129
aachild_Yes	0.033373	0.002399	-0.039310	-0.002669	0.030526	-0.018580	0.003292	0.013362	0.106912	0.069694	0.004222
dachild_Yes	0.005129	-0.011490	-0.065956	-0.011157	0.074157	-0.012426	0.013083	0.021575	0.201353	0.106122	-0.003291
childis_Yes	-0.003427	-0.010555	-0.023224	-0.014374	0.042810	0.012851	0.005272	-0.030453	-0.023235	-0.003415	0.145454
chbehprb_Yes	-0.023985	-0.026846	-0.007092	-0.012995	0.037128	0.006574	-0.006969	-0.024505	0.173573	0.278907	0.056225
prtsdied_Yes	-0.002156	0.019658	0.000717	0.022233	0.003905	-0.008476	-0.013928	0.006734	-0.006373	-0.009177	-0.005172
prtsjail_Yes	0.019518	-0.011959	-0.024845	-0.004040	0.052243	0.006558	0.010225	0.010143	-0.012998	-0.015838	-0.008829
abandmnt_Yes	0.011399	-0.001272	0.048078	0.001943	-0.025207	-0.018416	-0.026739	0.060924	0.003752	0.037627	0.016795
relinqsh_Yes	-0.024597	-0.004631	0.048471	-0.004269	-0.027085	-0.007801	-0.007872	-0.011210	-0.000406	-0.011155	0.011662
housing_Yes	0.004603	-0.013579	-0.012950	-0.004917	0.029378	0.009369	0.011296	0.060165	-0.029664	-0.046817	-0.008366
removal_2more	0.046290	-0.005692	0.036399	-0.000433	-0.021287	-0.007227	-0.009549	0.019273	0.028449	0.060493	0.076810
sex_x_female	0.016199	0.008046	-0.012369	0.009117	0.006174	0.005744	0.006083	0.037290	-0.033399	-0.177441	-0.028459

	clindis_Yes	mr_Yes	vishear_Yes	phydis_Yes	emotdist_Yes	othermed_Yes	manrem_Court ordered	curplset_Foster home, relative	curplset_Group home	curplset_Institution	curplset_Missing	curplset_Pre-adoptive home
amiakn_x_yes	0.017714	-0.000157	-0.011497	0.002261	0.012210	0.010636	0.007243	0.022361	-0.022031	-0.006838	-0.011240	0.034325
asian_x_yes	-0.013517	-0.017230	-0.000219	-0.010697	-0.022726	-0.002307	-0.017003	0.005487	-0.006706	-0.019005	-0.009550	-0.000027
blkafram_x_yes	0.001076	-0.014724	0.008942	-0.003106	0.016599	0.005112	-0.008043	-0.038484	0.007500	0.014959	-0.011755	-0.013680
hawaiipi_x_yes	-0.004768	-0.001471	0.014374	0.009316	-0.019455	-0.001003	0.003816	0.023977	-0.002105	-0.018299	-0.007022	0.011426
white_x_yes	0.004579	0.030246	-0.043474	0.007824	0.017505	-0.018503	0.006341	0.014723	0.001335	0.001361	0.026950	0.018063
currenroll_yes	-0.002859	-0.009910	-0.001768	0.004858	0.002197	0.010193	-0.014921	0.007382	0.002198	-0.024600	-0.029212	0.026692
cnctadult_yes	-0.011392	-0.011120	-0.002895	-0.013892	-0.003411	-0.011694	-0.000155	0.042815	-0.026354	-0.021576	0.013754	0.018600
homeless_yes	-0.013276	-0.030467	0.000090	-0.011572	-0.013621	-0.015084	0.016930	-0.038145	-0.000133	-0.000303	0.017843	-0.005976
subabuse_yes	-0.001512	-0.058543	-0.015082	-0.020277	0.013442	-0.033460	-0.004042	-0.072237	0.065731	0.118802	0.008137	-0.046908
incarc_yes	0.009839	-0.020815	-0.023641	-0.022555	0.050945	-0.036043	0.004841	-0.117678	0.096248	0.198043	0.022338	-0.057496
clindis_Yes	1.000000	0.204764	0.210740	0.108363	0.809009	0.471106	-0.059318	-0.103278	0.075325	0.035868	-0.005087	0.012221
mr_Yes	0.204764	1.000000	0.028409	0.109144	0.146676	0.150430	-0.025968	-0.051375	0.040326	0.034921	-0.004650	-0.000115
vishear_Yes	0.210740	0.028409	1.000000	0.045614	0.023060	0.126373	0.013170	-0.008174	0.017069	-0.015627	-0.016529	-0.016768
phydis_Yes	0.108363	0.109144	0.045614	1.000000	0.037352	0.082329	0.005922	-0.006292	0.018466	-0.018475	-0.007425	-0.004961
emotdist_Yes	0.809009	0.146676	0.023060	0.037352	1.000000	0.176670	-0.071360	-0.117016	0.069143	0.063620	0.000334	0.016176
othermed_Yes	0.471106	0.150430	0.126373	0.082329	0.176670	1.000000	-0.003903	-0.032684	0.043824	-0.005897	-0.013727	0.016954
manrem_Court ordered	-0.059318	-0.025968	0.013170	0.005922	-0.071360	-0.003903	1.000000	0.053289	-0.071430	-0.036448	0.016067	0.008190
curplset_Foster home, relative	-0.103278	-0.051375	-0.008174	-0.006292	-0.117016	-0.032684	0.053289	1.000000	-0.175178	-0.207656	-0.033568	-0.065400
curplset_Group home	0.075325	0.040326	0.017069	0.018466	0.069143	0.043824	-0.071430	-0.175178	1.000000	-0.214012	-0.034595	-0.067402
curplset_Institution	0.035868	0.034921	-0.015627	-0.018475	0.063620	-0.005897	-0.036448	-0.207656	-0.214012	1.000000	-0.041009	-0.079899
curplset_Missing	-0.005087	-0.004650	-0.016529	-0.007425	0.000334	-0.013727	0.016067	-0.033568	-0.034595	-0.041009	1.000000	-0.012916
curplset_Pre-adoptive home	0.012221	-0.000115	-0.016768	-0.004961	0.016176	0.016954	0.008190	-0.065400	-0.067402	-0.079899	-0.012916	1.000000
curplset_Runaway	0.007696	-0.013193	0.004947	-0.003168	0.010074	-0.008488	0.004992	-0.083033	-0.085575	-0.101441	-0.016398	-0.031948
curplset_Supervised independent living	0.007191	0.011522	0.004666	0.010197	0.014341	-0.010148	-0.006916	-0.063839	-0.065793	-0.077991	-0.012607	-0.024563
curplset_Trial home visit	-0.036127	-0.013394	-0.024336	0.000469	-0.026382	-0.019867	0.011269	-0.091860	-0.094672	-0.112224	-0.018141	-0.035344
placeout_Yes	0.024116	0.013368	-0.008913	-0.000409	0.028685	0.010574	0.002294	0.020900	-0.003108	0.081891	0.017143	0.035587
phyabuse_Yes	-0.001815	0.021628	-0.012297	-0.006268	-0.006615	0.014779	0.058427	0.018612	-0.047487	-0.026644	0.018104	0.015255
sexabuse_Yes	-0.008222	0.012796	-0.001361	0.001476	-0.002828	-0.000123	0.029187	-0.005387	-0.021702	-0.013225	0.025654	0.027747
neglect_Yes	-0.028497	0.007221	0.054455	0.011614	-0.048630	0.015799	0.155935	0.103369	-0.103845	-0.079830	0.043123	0.035395
aaparent_Yes	-0.026027	-0.005092	-0.010020	0.005687	-0.026774	-0.002283	0.025987	0.056835	-0.047440	-0.039643	-0.001314	0.020762
daparent_Yes	-0.048129	-0.011414	-0.032735	0.006169	-0.046031	-0.018736	0.062846	0.130655	-0.070301	-0.074267	0.038971	0.051223
aachild_Yes	0.004222	-0.009149	-0.010470	-0.003600	0.017885	-0.005242	0.006079	-0.017317	-0.002665	0.007708	0.005702	-0.013925
dachild_Yes	-0.003291	-0.015377	-0.020849	-0.001961	0.002710	0.001250	0.010024	-0.017138	0.010795	0.039085	-0.001718	-0.017711
childis_Yes	0.145454	0.136656	-0.000692	0.068037	0.130864	0.097581	-0.093102	-0.040398	0.012631	0.075410	0.027545	-0.007991
chbehprb_Yes	0.056225	0.021534	-0.054005	-0.012607	0.100767	-0.012361	-0.085704	-0.161012	0.133676	0.174737	-0.040366	-0.058918
prtsdied_Yes	-0.005172	0.001752	-0.010176	0.001349	-0.007818	0.003135	0.007129	0.018432	-0.012476	-0.026030	-0.009831	0.020584
prtsjail_Yes	-0.008829	0.008167	-0.011862	0.018699	-0.012392	0.008380	0.033438	0.029700	-0.046283	-0.020920	0.025416	0.008403
abandmnt_Yes	0.016795	-0.002057	-0.037630	0.001066	0.032911	-0.019708	0.053106	-0.024563	-0.042823	0.048735	0.059882	-0.009095
relinqsh_Yes	0.011662	0.005792	-0.007299	-0.000967	0.020547	-0.007425	-0.048730	-0.026769	0.004086	0.006553	-0.013270	-0.017731
housing_Yes	-0.008366	0.023494	-0.015845	0.004405	-0.011201	0.007462	0.034327	0.013087	-0.039791	-0.017850	0.030040	0.016585
removal_2more	0.076810	0.007486	0.024461	0.008612	0.068541	0.043267	-0.028074	-0.047283	0.015353	0.030444	-0.009367	-0.020666
sex_x_female	-0.028459	-0.050481	0.002548	0.000255	-0.040832	-0.018775	0.002166	0.027127	-0.076317	-0.079451	-0.003223	0.003495

A	curplset_Pre-adoptive home (W)	curplset_Runaway (X)	curplset_Supervised independent living (Y)	curplset_Trial home visit (Z)	placeout_Yes (AA)	phyabuse_Yes (AB)	sexabuse_Yes (AC)	neglect_Yes (AD)	aaparent_Yes (AE)
amiakn_x_yes	0.034325	0.026050	-0.020322	-0.010803	0.016531	-0.011402	-0.008032	0.009589	0.071059
asian_x_yes	-0.000027	-0.005799	0.008453	-0.001756	-0.002394	0.011938	0.012050	-0.002907	-0.002575
blkafram_x_yes	-0.013680	0.017026	0.018376	-0.021327	-0.001011	0.034747	-0.065413	-0.025354	-0.041616
hawaiipi_x_yes	0.011426	-0.005327	-0.013355	-0.008235	-0.007256	0.018191	0.004623	0.000468	-0.003235
white_x_yes	0.018063	-0.038353	-0.014575	0.026953	0.008421	-0.020680	0.064469	0.010357	0.023210
currenroll_yes	0.026692	-0.064574	-0.002316	0.002213	-0.017766	0.007462	0.006268	-0.010479	-0.005074
cnctadult_yes	0.018600	-0.019562	-0.030850	0.006563	0.012569	0.010999	0.015628	0.003874	0.010548
homeless_yes	-0.005976	0.068825	0.008517	-0.037205	0.008182	-0.033895	-0.029819	0.047470	-0.002474
subabuse_yes	-0.046908	0.079877	0.006664	0.046409	0.029309	-0.056583	-0.058326	-0.087821	-0.005158
incarc_yes	-0.057496	0.091596	-0.002051	0.035359	0.051667	-0.069115	-0.070196	-0.151927	-0.032156
clindis_Yes	0.012221	0.007696	0.007191	-0.036127	0.024116	-0.001815	-0.008222	-0.028497	-0.026027
mr_Yes	-0.000115	-0.013193	0.011522	-0.013394	0.013368	0.021628	0.012796	0.007221	-0.005092
vishear_Yes	-0.016768	0.004947	0.004666	-0.024336	-0.008913	-0.012297	-0.001361	0.054455	-0.010020
phydis_Yes	-0.004961	-0.003168	0.010197	0.000469	-0.000409	-0.006268	0.001476	0.011614	0.005687
emotdist_Yes	0.016176	0.010074	0.014341	-0.026382	0.028685	-0.006615	-0.002828	-0.048630	-0.026774
othermed_Yes	0.016954	-0.008488	-0.010148	-0.019867	0.010574	0.014779	-0.000123	0.015799	-0.002283
manrem_Court ordered	0.008190	0.004992	-0.006916	0.011269	0.002294	0.058427	0.029187	0.155935	0.025987
curplset_Foster home, relative	-0.065400	-0.083033	-0.063839	-0.091860	0.020900	0.018612	-0.005387	0.103369	0.056835
curplset_Group home	-0.067402	-0.085575	-0.065793	-0.094672	-0.003108	-0.047487	-0.021702	-0.103845	-0.047440
curplset_Institution	-0.079899	-0.101441	-0.077991	-0.112224	0.081891	-0.026644	-0.013225	-0.079830	-0.039643
curplset_Missing	-0.012916	-0.016398	-0.012607	-0.018141	0.017143	0.018104	0.025654	0.043123	-0.001314
curplset_Pre-adoptive home	1.000000	-0.031948	-0.024563	-0.035344	0.035587	0.015255	0.027747	0.035395	0.020762
curplset_Runaway	-0.031948	1.000000	-0.031186	-0.044874	-0.031474	0.004239	-0.015284	0.000845	-0.009432
curplset_Supervised independent living	-0.024563	-0.031186	1.000000	-0.034501	-0.023711	-0.009198	-0.005772	-0.036480	-0.005129
curplset_Trial home visit	-0.035344	-0.044874	-0.034501	1.000000	-0.000887	-0.017699	-0.017916	-0.045988	0.024413
placeout_Yes	0.035587	-0.031474	-0.023711	-0.000887	1.000000	0.007233	-0.002485	-0.025237	0.009607
phyabuse_Yes	0.015255	0.004239	-0.009198	-0.017699	0.007233	1.000000	0.052537	0.014892	0.053649
sexabuse_Yes	0.027747	-0.015284	-0.005772	-0.017916	-0.002485	0.052537	1.000000	-0.001372	0.001958
neglect_Yes	0.035395	0.000845	-0.036480	-0.045988	-0.025237	0.014892	-0.001372	1.000000	0.088021
aaparent_Yes	0.020762	-0.009432	-0.005129	0.024413	0.009607	0.053649	0.001958	0.088021	1.000000
daparent_Yes	0.051223	-0.005826	-0.026230	0.009004	-0.021110	0.004606	-0.024075	0.159718	0.163568
aachild_Yes	-0.013925	0.030111	-0.009441	0.020232	0.002745	-0.003408	0.009189	-0.001381	0.105180
dachild_Yes	-0.017711	0.037836	0.002641	0.041173	0.003149	-0.014453	-0.013525	-0.027437	0.046555
childis_Yes	-0.007991	-0.008543	-0.021827	-0.019163	0.008786	0.040366	0.024800	0.001733	0.001516
chbehprb_Yes	-0.058918	0.032276	0.040849	0.074785	0.029145	-0.163187	-0.091982	-0.374964	-0.059861
prtsdied_Yes	0.020584	-0.012765	0.014573	-0.024268	-0.006694	-0.012728	-0.006422	-0.034035	0.004265
prtsjail_Yes	0.008403	-0.001410	-0.002383	0.003616	-0.006723	-0.000042	-0.000027	0.032750	0.069844
abandmnt_Yes	-0.009095	0.030334	-0.005338	-0.019766	0.005732	-0.033425	-0.034341	-0.029768	-0.014624
relinqsh_Yes	-0.017731	0.001811	0.027407	-0.010655	-0.006703	-0.033519	-0.025285	-0.086900	-0.022546
housing_Yes	0.016585	-0.001513	-0.005390	-0.010756	0.001445	-0.013802	-0.013903	0.124841	0.072197
removal_2more	-0.020666	0.033390	0.009387	-0.019558	-0.006632	-0.022752	-0.052901	-0.042287	-0.012231
sex_x_female	0.003495	0.007816	0.010398	0.000015	-0.019968	0.044331	0.133404	0.068379	0.010950

A	daparent_Yes (AF)	aachild_Yes (AG)	dachild_Yes (AH)	childis_Yes (AI)	chbehprb_Yes (AJ)	prtsdied_Yes (AK)	prtsjail_Yes (AL)	abandmnt_Yes (AM)	relinqsh_Yes (AN)	housing_Yes (AO)	removal_2more (AP)	sex_x_female (AQ)
amiakn_x_yes	0.015500	0.033373	0.005129	-0.003427	-0.023985	-0.002156	0.019518	0.011399	-0.024597	0.004603	0.046290	0.016199
asian_x_yes	-0.018580	0.002399	-0.011490	-0.010555	-0.026846	0.019658	-0.011959	-0.001272	-0.004631	-0.013579	-0.005692	0.008046
blkafram_x_yes	-0.090842	-0.039310	-0.065956	-0.023224	-0.007092	0.000717	-0.024845	0.048078	0.048471	-0.012950	0.036399	-0.012369
hawaiipi_x_yes	-0.000707	-0.002669	-0.011157	-0.014374	-0.012995	0.022233	-0.004040	0.001943	-0.004269	-0.004917	-0.000433	0.009117
white_x_yes	0.108449	0.030526	0.074157	0.042810	0.037128	0.003905	0.052243	-0.025207	-0.027085	0.029378	-0.021287	0.006174
currenroll_yes	-0.008105	-0.018580	-0.012426	0.012851	0.006574	-0.008476	0.006558	-0.018416	-0.007801	0.009369	-0.007227	0.005744
cnctadult_yes	0.022181	0.003292	0.013083	0.005272	-0.006969	-0.013928	0.010225	-0.026739	-0.007872	0.011296	-0.009549	0.006083
homeless_yes	0.020103	0.013362	0.021575	-0.030453	-0.024505	0.006734	0.010143	0.060924	-0.011210	0.060165	0.019273	0.037290
subabuse_yes	-0.016729	0.106912	0.201353	-0.023235	0.173573	-0.006373	-0.012998	0.003752	-0.000406	-0.029664	0.028449	-0.033399
incarc_yes	-0.078161	0.069694	0.106122	-0.003415	0.278907	-0.009177	-0.015838	0.037627	-0.011155	-0.046817	0.060493	-0.177441
clindis_Yes	-0.048129	0.004222	-0.003291	0.145454	0.056225	-0.005172	-0.008829	0.016795	0.011662	-0.008366	0.076810	-0.028459
mr_Yes	-0.011414	-0.009149	-0.015377	0.136656	0.021534	0.001752	0.008167	-0.002057	0.005792	0.023494	0.007486	-0.050481
vishear_Yes	-0.032735	-0.010470	-0.020849	-0.000692	-0.054005	-0.010176	-0.011862	-0.037630	-0.007299	-0.015845	0.024461	0.002548
phydis_Yes	0.006169	-0.003600	-0.001961	0.068037	-0.012607	0.001349	0.018699	0.001066	-0.000967	0.004405	0.008612	0.000255
emotdist_Yes	-0.046031	0.017885	0.002710	0.130864	0.100767	-0.007818	-0.012392	0.032911	0.020547	-0.011201	0.068541	-0.040832
othermed_Yes	-0.018736	-0.005242	0.001250	0.097581	-0.012361	0.003135	0.008380	-0.019708	-0.007425	0.007462	0.043267	-0.018775
manrem_Court ordered	0.062846	0.006079	0.010024	-0.093102	-0.085704	0.007129	0.033438	0.053106	-0.048730	0.034327	-0.028074	0.002166
curplset_Foster home, relative	0.130655	-0.017317	-0.017138	-0.040398	-0.161012	0.018432	0.029700	-0.024563	-0.026769	0.013087	-0.047283	0.027127
curplset_Group home	-0.070301	-0.002665	0.010795	0.012631	0.133676	-0.012476	-0.046283	-0.042823	0.004086	-0.039791	0.015353	-0.076317
curplset_Institution	-0.074267	0.007708	0.039085	0.075410	0.174737	-0.026030	-0.020920	0.048735	0.006553	-0.017850	0.030444	-0.079451
curplset_Missing	0.038971	0.005702	-0.001718	0.027545	-0.040366	-0.009831	0.025416	0.059882	-0.013270	0.030040	-0.009367	-0.003223
curplset_Pre-adoptive home	0.051223	-0.013925	-0.017711	-0.007991	-0.058918	0.020584	0.008403	-0.009095	-0.017731	0.016585	-0.020666	0.003495
curplset_Runaway	-0.005826	0.030111	0.037836	-0.008543	0.032276	-0.012765	-0.001410	0.030334	0.001811	-0.001513	0.033390	0.007816
curplset_Supervised independent living	-0.026230	-0.009441	0.002641	-0.021827	0.040849	0.014573	-0.002383	-0.005338	0.027407	-0.005390	0.009387	0.010398
curplset_Trial home visit	0.009004	0.020232	0.041173	-0.019163	0.074785	-0.024268	0.003616	-0.019766	-0.010655	-0.010756	-0.019558	0.000015
placeout_Yes	-0.021110	0.002745	0.003149	0.008786	0.029145	-0.006694	-0.006723	0.005732	-0.006703	0.001445	-0.006632	-0.019968
phyabuse_Yes	0.004606	-0.003408	-0.014453	0.040366	-0.163187	-0.012728	-0.000042	-0.033425	-0.033519	-0.013802	-0.022752	0.044331
sexabuse_Yes	-0.024075	0.009189	-0.013525	0.024800	-0.091982	-0.006422	-0.000027	-0.034341	-0.025285	-0.013903	-0.052901	0.133404
neglect_Yes	0.159718	-0.001381	-0.027437	0.001733	-0.374964	-0.034035	0.032750	-0.029768	-0.086900	0.124841	-0.042287	0.068379
aaparent_Yes	0.163568	0.105180	0.046555	0.001516	-0.059861	0.004265	0.069844	-0.014624	-0.022546	0.072197	-0.012231	0.010950
daparent_Yes	1.000000	0.026454	0.073829	0.002316	-0.178348	0.000324	0.118935	-0.039710	-0.045764	0.161969	-0.037647	0.025257
aachild_Yes	0.026454	1.000000	0.370410	0.047534	0.089424	0.016709	0.006972	0.012044	0.004389	0.019489	-0.001394	0.003296
dachild_Yes	0.073829	0.370410	1.000000	0.060867	0.151945	0.011704	0.023311	0.005136	0.007264	0.020297	0.007643	-0.002842
childis_Yes	0.002316	0.047534	0.060867	1.000000	0.083625	-0.000621	0.005625	0.100265	0.000803	0.022221	0.001593	-0.026633
chbehprb_Yes	-0.178348	0.089424	0.151945	0.083625	1.000000	-0.016382	-0.072350	-0.041241	0.035377	-0.085676	0.055298	-0.113303
prtsdied_Yes	0.000324	0.016709	0.011704	-0.000621	-0.016382	1.000000	0.026997	0.016889	0.001446	0.000476	0.002904	-0.000204
prtsjail_Yes	0.118935	0.006972	0.023311	0.005625	-0.072350	0.026997	1.000000	-0.004009	-0.013682	0.069080	-0.011587	0.007859
abandmnt_Yes	-0.039710	0.012044	0.005136	0.100265	-0.041241	0.016889	-0.004009	1.000000	0.009151	0.025974	-0.006363	-0.017618
relinqsh_Yes	-0.045764	0.004389	0.007264	0.000803	0.035377	0.001446	-0.013682	0.009151	1.000000	-0.017632	0.034770	0.001482
housing_Yes	0.161969	0.019489	0.020297	0.022221	-0.085676	0.000476	0.069080	0.025974	-0.017632	1.000000	-0.033788	0.002814
removal_2more	-0.037647	-0.001394	0.007643	0.001593	0.055298	0.002904	-0.011587	-0.006363	0.034770	-0.033788	1.000000	-0.011312
sex_x_female	0.025257	0.003296	-0.002842	-0.026633	-0.113303	-0.000204	0.007859	-0.017618	0.001482	0.002814	-0.011312	1.000000

	daparent_Yes	aachild_Yes	dachild_Yes	childis_Yes	chbehprb_Yes	prtsdied_Yes	prtsjail_Yes	abandmnt_Yes	relinqsh_Yes	housing_Yes	removal_2more	sex_x_female
amiakn_x_yes	0.015500	0.033373	0.005129	-0.003427	-0.023985	-0.002156	0.019518	0.011399	-0.024597	0.004603	0.046290	0.016199
asian_x_yes	-0.018580	0.002399	-0.011490	-0.010555	-0.026846	0.019658	-0.011959	-0.001272	-0.004631	-0.013579	-0.005692	0.008046
blkafram_x_yes	-0.090842	-0.039310	-0.065956	-0.023224	-0.007092	0.000717	-0.024845	0.048078	0.048471	-0.012950	0.036399	-0.012369
hawaiipi_x_yes	-0.000707	-0.002669	-0.011157	-0.014374	-0.012995	0.022233	-0.004040	0.001943	-0.004269	-0.004917	-0.000433	0.009117
white_x_yes	0.108449	0.030526	0.074157	0.042810	0.037128	0.003905	0.052243	-0.025207	-0.027085	0.029378	-0.021287	0.006174
currenroll_yes	-0.008105	-0.018580	-0.012426	0.012851	0.006574	-0.008476	0.006558	-0.018416	-0.007801	0.009369	-0.007227	0.005744
cnctadult_yes	0.022181	0.003292	0.013083	0.005272	-0.006969	-0.013928	0.010225	-0.026739	-0.007872	0.011296	-0.009549	0.006083
homeless_yes	0.020103	0.013362	0.021575	-0.030453	-0.024505	0.006734	0.010143	0.060924	-0.011210	0.060165	0.019273	0.037290
subabuse_yes	-0.016729	0.106912	0.201353	-0.023235	0.173573	-0.006373	-0.012998	0.003752	-0.000406	-0.029664	0.028449	-0.033399
incarc_yes	-0.078161	0.069694	0.106122	-0.003415	0.278907	-0.009177	-0.015838	0.037627	-0.011155	-0.046817	0.060493	-0.177441
clindis_Yes	-0.048129	0.004222	-0.003291	0.145454	0.056225	-0.005172	-0.008829	0.016795	0.011662	-0.008366	0.076810	-0.028459
mr_Yes	-0.011414	-0.009149	-0.015377	0.136656	0.021534	0.001752	0.008167	-0.002057	0.005792	0.023494	0.007486	-0.050481
vishear_Yes	-0.032735	-0.010470	-0.020849	-0.000692	-0.054005	-0.010176	-0.011862	-0.037630	-0.007299	-0.015845	0.024461	0.002548
phydis_Yes	0.006169	-0.003600	-0.001961	0.068037	-0.012607	0.001349	0.018699	0.001066	-0.000967	0.004405	0.008612	0.000255
emotdist_Yes	-0.046031	0.017885	0.002710	0.130864	0.100767	-0.007818	-0.012392	0.032911	0.020547	-0.011201	0.068541	-0.040832
othermed_Yes	-0.018736	-0.005242	0.001250	0.097581	-0.012361	0.003135	0.008380	-0.019708	-0.007425	0.007462	0.043267	-0.018775
manrem_Court ordered	0.062846	0.006079	0.010024	-0.093102	-0.085704	0.007129	0.033438	0.053106	-0.048730	0.034327	-0.028074	0.002166
curplset_Foster home, relative	0.130655	-0.017317	-0.017138	-0.040398	-0.161012	0.018432	0.029700	-0.024563	-0.026769	0.013087	-0.047283	0.027127
curplset_Group home	-0.070301	-0.002665	0.010795	0.012631	0.133676	-0.012476	-0.046283	-0.042823	0.004086	-0.039791	0.015353	-0.076317
curplset_Institution	-0.074267	0.007708	0.039085	0.075410	0.174737	-0.026030	-0.020920	0.048735	0.006553	-0.017850	0.030444	-0.079451
curplset_Missing	0.038971	0.005702	-0.001718	0.027545	-0.040366	-0.009831	0.025416	0.059882	-0.013270	0.030040	-0.009367	-0.003223
curplset_Pre-adoptive home	0.051223	-0.013925	-0.017711	-0.007991	-0.058918	0.020584	0.008403	-0.009095	-0.017731	0.016585	-0.020666	0.003495
curplset_Runaway	-0.005826	0.030111	0.037836	-0.008543	0.032276	-0.012765	-0.001410	0.030334	0.001811	-0.001513	0.033390	0.007816
curplset_Supervised independent living	-0.026230	-0.009441	0.002641	-0.021827	0.040849	0.014573	-0.002383	-0.005338	0.027407	-0.005390	0.009387	0.010398
curplset_Trial home visit	0.009004	0.020232	0.041173	-0.019163	0.074785	-0.024268	0.003616	-0.019766	-0.010655	-0.010756	-0.019558	0.000015
placeout_Yes	-0.021110	0.002745	0.003149	0.008786	0.029145	-0.006694	-0.006723	0.005732	-0.006703	0.001445	-0.006632	-0.019968
phyabuse_Yes	0.004606	-0.003408	-0.014453	0.040366	-0.163187	-0.012728	-0.000042	-0.033425	-0.033519	-0.013802	-0.022752	0.044331
sexabuse_Yes	-0.024075	0.009189	-0.013525	0.024800	-0.091982	-0.006422	-0.000027	-0.034341	-0.025285	-0.013903	-0.052901	0.133404
neglect_Yes	0.159718	-0.001381	-0.027437	0.001733	-0.374964	-0.034035	0.032750	-0.029768	-0.086900	0.124841	-0.042287	0.068379
aaparent_Yes	0.163568	0.105180	0.046555	0.001516	-0.059861	0.004265	0.069844	-0.014624	-0.022546	0.072197	-0.012231	0.010950
daparent_Yes	1.000000	0.026454	0.073829	0.002316	-0.178348	0.000324	0.118935	-0.039710	-0.045764	0.161969	-0.037647	0.025257
aachild_Yes	0.026454	1.000000	0.370410	0.047534	0.089424	0.016709	0.006972	0.012044	0.004389	0.019489	-0.001394	0.003296
dachild_Yes	0.073829	0.370410	1.000000	0.060867	0.151945	0.011704	0.023311	0.005136	0.007264	0.020297	0.007643	-0.002842
childis_Yes	0.002316	0.047534	0.060867	1.000000	0.083625	-0.000621	0.005625	0.100265	0.000803	0.022221	0.001593	-0.026633
chbehprb_Yes	-0.178348	0.089424	0.151945	0.083625	1.000000	-0.016382	-0.072350	-0.041241	0.035377	-0.085676	0.055298	-0.113303
prtsdied_Yes	0.000324	0.016709	0.011704	-0.000621	-0.016382	1.000000	0.026997	0.016889	0.001446	0.000476	0.002904	-0.000204
prtsjail_Yes	0.118935	0.006972	0.023311	0.005625	-0.072350	0.026997	1.000000	-0.004009	-0.013682	0.069080	-0.011587	0.007859
abandmnt_Yes	-0.039710	0.012044	0.005136	0.100265	-0.041241	0.016889	-0.004009	1.000000	0.009151	0.025974	-0.006363	-0.017618
relinqsh_Yes	-0.045764	0.004389	0.007264	0.000803	0.035377	0.001446	-0.013682	0.009151	1.000000	-0.017632	0.034770	0.001482
housing_Yes	0.161969	0.019489	0.020297	0.022221	-0.085676	0.000476	0.069080	0.025974	-0.017632	1.000000	-0.033788	0.002814
removal_2more	-0.037647	-0.001394	0.007643	0.001593	0.055298	0.002904	-0.011587	-0.006363	0.034770	-0.033788	1.000000	-0.011312
sex_x_female	0.025257	0.003296	-0.002842	-0.026633	-0.113303	-0.000204	0.007859	-0.017618	0.001482	0.002814	-0.011312	1.000000

Appendix C. Incarceration Response Distribution per State

Rank	State	incarc_yes	incarc_no	incarc_blank	incarc_declined	Yes/ No
1	CO	56.79%	26.13%	14.63%	2.44%	2.17
2	ID	51.16%	45.35%	2.33%	1.16%	1.13
3	WY	43.59%	52.56%	3.85%	0.00%	0.83
4	UT	41.52%	37.05%	19.64%	1.79%	1.12
5	WI	40.44%	50.55%	8.20%	0.82%	0.80
6	VA	35.19%	54.57%	9.80%	0.45%	0.64
7	ND	34.86%	33.03%	32.11%	0.00%	1.06
8	DE	33.90%	49.15%	16.95%	0.00%	0.69
9	SD	33.33%	61.67%	3.33%	1.67%	0.54
10	LA	33.02%	58.14%	7.91%	0.93%	0.57

11	IA	32.31%	54.23%	11.92%	1.54%	0.60
12	RI	32.00%	52.80%	15.20%	0.00%	0.61
13	NM	31.51%	68.49%	0.00%	0.00%	0.46
14	WA	31.41%	50.69%	16.80%	1.10%	0.62
15	IN	30.99%	52.67%	14.96%	1.37%	0.59
16	MN	30.79%	46.07%	22.05%	1.09%	0.67
17	NV	29.41%	57.35%	13.24%	0.00%	0.51
18	DC	29.17%	56.25%	14.58%	0.00%	0.52
19	TX	26.77%	55.76%	16.73%	0.74%	0.48
20	TN	26.53%	26.05%	47.27%	0.16%	1.02
21	SC	24.91%	51.54%	22.53%	1.02%	0.48
22	AR	23.89%	46.02%	28.76%	1.33%	0.52
23	MT	22.77%	53.47%	21.78%	1.98%	0.43
24	OH	22.43%	24.84%	46.62%	6.12%	0.90
25	GA	21.81%	61.19%	16.15%	0.85%	0.36
26	KS	21.55%	64.66%	12.03%	1.75%	0.33
27	IL	20.42%	55.80%	21.96%	1.82%	0.37
28	KY	19.75%	56.40%	20.85%	3.00%	0.35
29	NE	19.55%	62.57%	11.73%	6.15%	0.31
30	PA	19.08%	56.66%	20.71%	3.55%	0.34
31	CA	17.11%	51.41%	30.70%	0.78%	0.33
32	WV	16.96%	48.25%	29.53%	5.26%	0.35
33	ME	16.95%	62.71%	20.34%	0.00%	0.27
34	HI	16.88%	62.34%	19.48%	1.30%	0.27
35	VT	16.13%	83.87%	0.00%	0.00%	0.19
36	NH	15.96%	54.26%	29.79%	0.00%	0.29
37	CT	15.84%	76.02%	8.14%	0.00%	0.21
38	NY	15.75%	42.02%	41.24%	0.99%	0.37
39	OR	15.26%	43.61%	36.45%	4.67%	0.35
40	OK	14.54%	71.37%	12.33%	1.76%	0.20
41	AL	14.45%	62.24%	22.12%	1.18%	0.23
42	MA	13.83%	60.81%	23.76%	1.61%	0.23
43	MO	13.16%	53.60%	32.13%	1.11%	0.25
44	AK	12.66%	56.96%	30.38%	0.00%	0.22
45	NJ	8.76%	81.67%	9.56%	0.00%	0.11

46	FL	7.76%	18.92%	72.66%	0.66%	0.41
47	NC	7.51%	22.18%	69.11%	1.19%	0.34
48	MD	7.12%	77.74%	14.84%	0.30%	0.09
49	MS	5.13%	52.31%	42.56%	0.00%	0.10
50	AZ	4.69%	11.02%	83.94%	0.35%	0.43
51	PR	2.13%	97.87%	0.00%	0.00%	0.02
52	MI	0.00%	0.00%	14.98%	85.02%	n/a

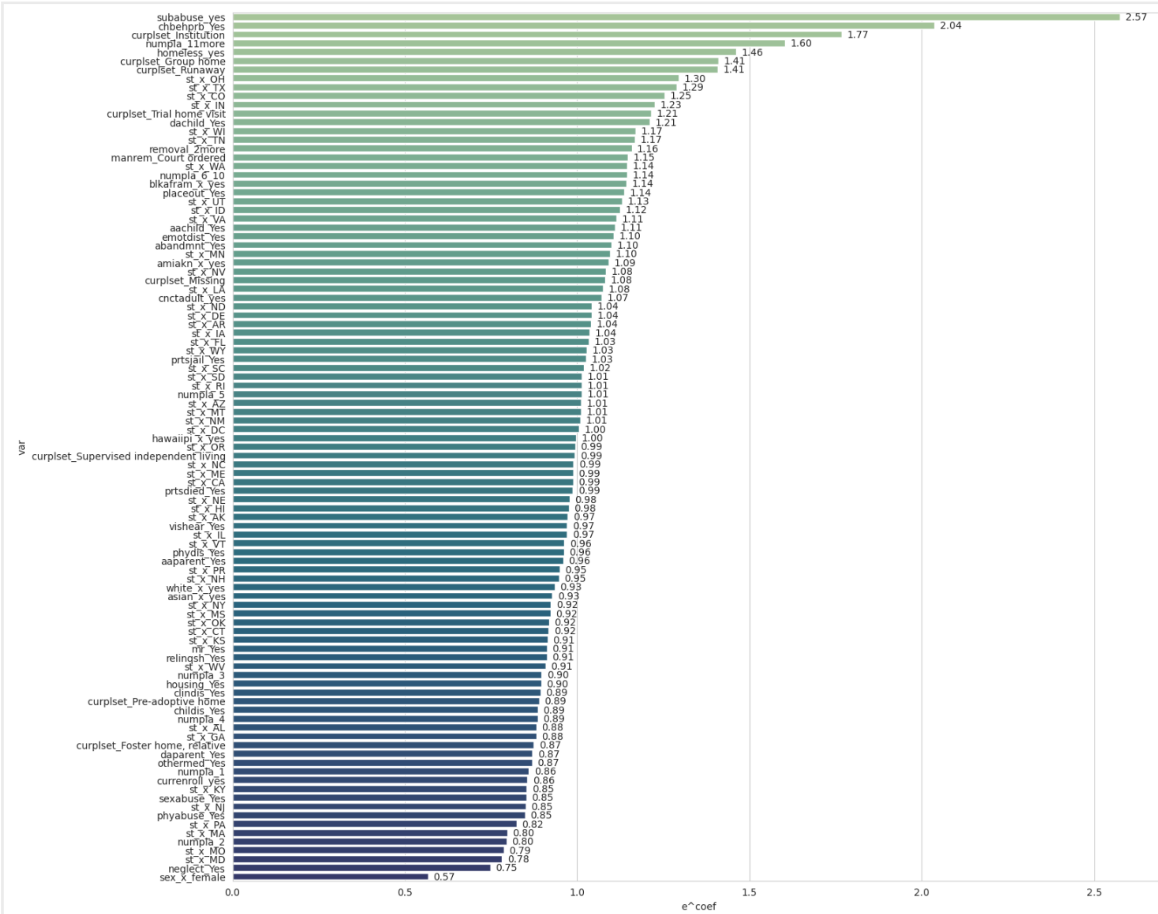
Appendix D. Data Set Distribution Post the Implementation of SMOTE-ENN

variable	count	count (post smote)	percentage	percentage (post smote)
sex_x_female	7738	5351	5.35%	4.82%
amiakn_x_yes	537	320	0.37%	0.29%
asian_x_yes	204	117	0.14%	0.11%
blkafram_x_yes	4969	3568	3.43%	3.21%
hawaiipi_x_yes	111	72	0.08%	0.06%
white_x_yes	9439	7226	6.52%	6.51%

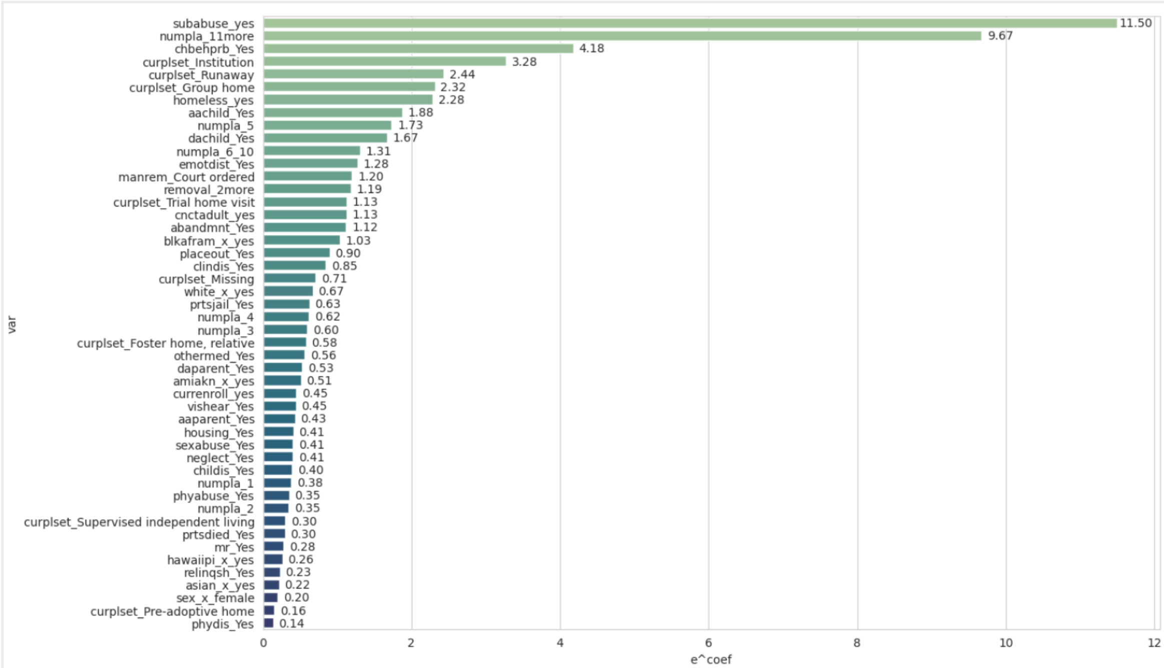
currenroll_yes	13980	10785	9.66%	9.72%
cnctadult_yes	14124	11012	9.76%	9.92%
homeless_yes	3206	2479	2.22%	2.23%
subabuse_yes	3610	3547	2.49%	3.20%
clindis_Yes	6233	4540	4.31%	4.09%
mr_Yes	450	213	0.31%	0.19%
vishear_Yes	595	364	0.41%	0.33%
phydis_Yes	124	61	0.09%	0.05%
emotdist_Yes	4982	3621	3.44%	3.26%
othermed_Yes	2092	1402	1.45%	1.26%
manrem_Court ordered	14204	11072	9.82%	9.98%
curplset_Foster home, relative	2184	1489	1.51%	1.34%
curplset_Group home	2299	1704	1.59%	1.54%
curplset_Institution	3042	2636	2.10%	2.37%
curplset_Missing	99	49	0.07%	0.04%
curplset_Pre-adoptive home	369	241	0.26%	0.22%
curplset_Runaway	586	401	0.40%	0.36%
curplset_Supervised independent living	352	182	0.24%	0.16%
curplset_Trial home visit	711	431	0.49%	0.39%

placeout_Yes	497	336	0.34%	0.30%
phyabuse_Yes	1921	1165	1.33%	1.05%
sexabuse_Yes	1266	779	0.87%	0.70%
neglect_Yes	7408	5138	5.12%	4.63%
aaparent_Yes	657	354	0.45%	0.32%
daparent_Yes	2074	1296	1.43%	1.17%
aachild_Yes	187	134	0.13%	0.12%
dachild_Yes	673	467	0.47%	0.42%
childis_Yes	642	330	0.44%	0.30%
chbehprb_Yes	4851	4095	3.35%	3.69%
prtsdied_Yes	216	103	0.15%	0.09%
prtsjail_Yes	689	365	0.48%	0.33%
abandmnt_Yes	1523	957	1.05%	0.86%
relinqsh_Yes	389	180	0.27%	0.16%
housing_Yes	1029	547	0.71%	0.49%
numpla_1	2921	2492.46737	2.02%	2.25%
numpla_2	2719	1985.623978	1.88%	1.79%
numpla_3	2082	1628.830849	1.44%	1.47%
numpla_4	1512	1111.391226	1.04%	1.00%
numpla_5	1082	791.798502	0.75%	0.71%
numpla_6_10	2798	2092.326916	1.93%	1.89%
numpla_11more	1915	1599.561159	1.32%	1.44%
removal_2more	4911	3531	3.39%	3.18%
incarc_yes	4490	6631	3.10%	5.97%

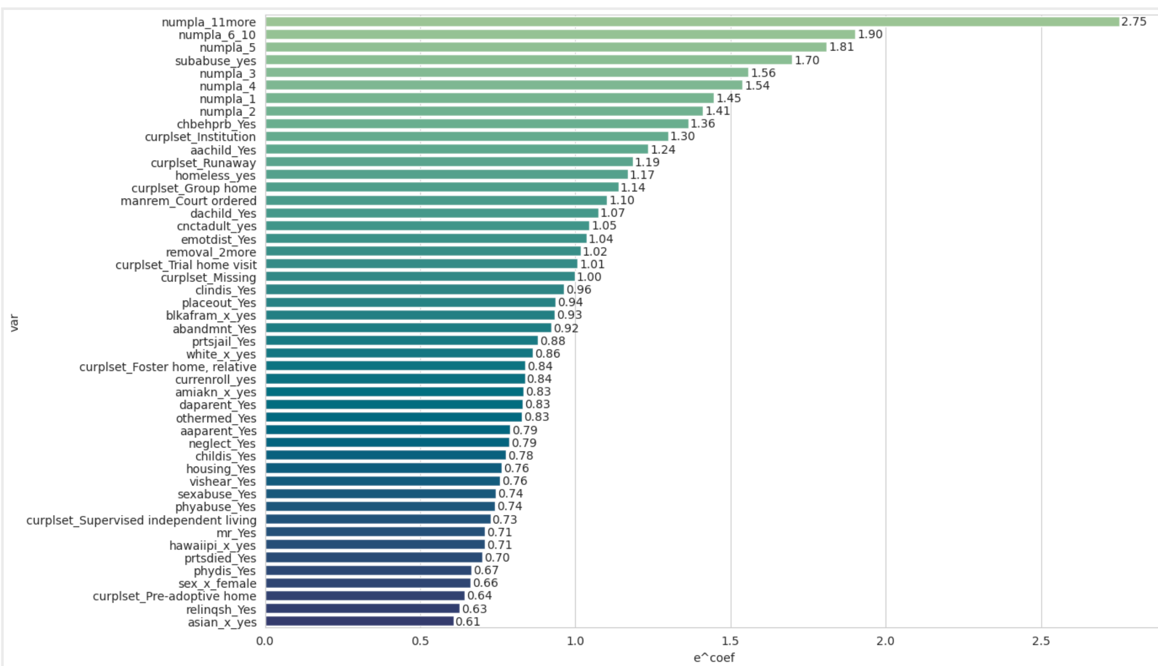
Appendix E. Baseline Logistic Regression Plot



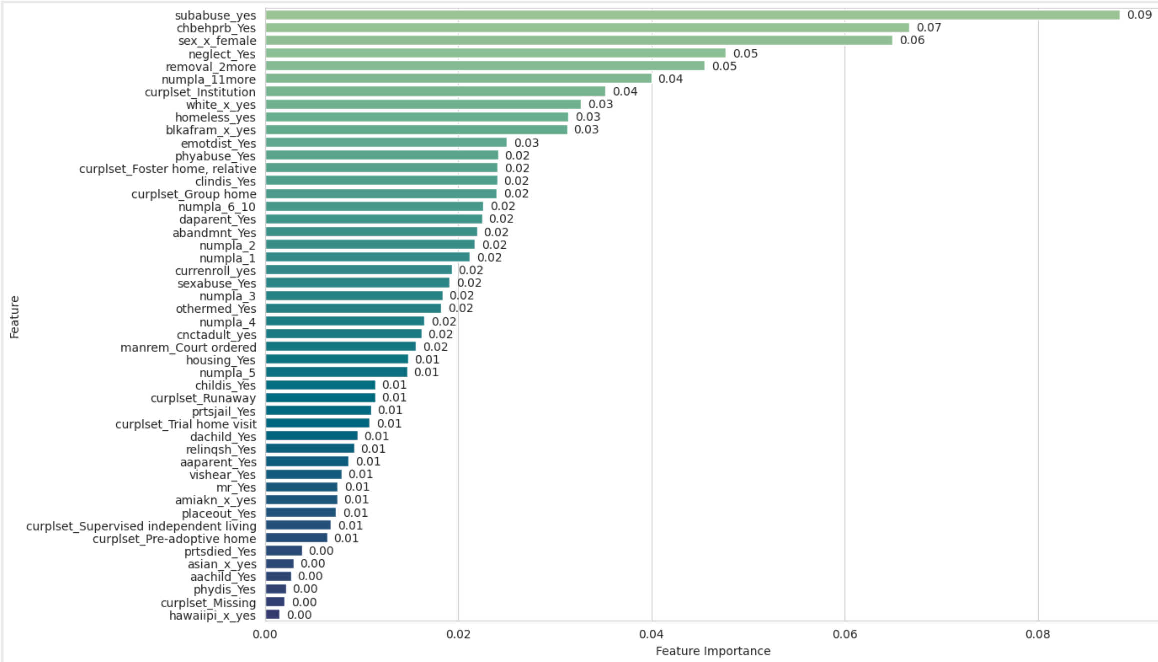
Appendix F. Smote-ENN Logistic Regression without States variable importance plot



Appendix G. SVM variable importance plot



Appendix H. Random Forest variable importance plot


subabuse_yes 0.09
chbehprb_Yes 0.07
sex_x_female 0.06
neglect_Yes 0.05
removal_2more 0.05
numpla_11more 0.04
curplset_Institution 0.04
white_x_yes 0.03
homeless_yes 0.03
blkafram_x_yes 0.03
emotdist_Yes 0.03
phyabuse_Yes 0.02
curplset_Foster home, relative 0.02
clindis_Yes 0.02
curplset_Group home 0.02
numpla_6_10 0.02
daparent_Yes 0.02
abandmnt_Yes 0.02
numpla_2 0.02
numpla_1 0.02
currenroll_yes 0.02
sexabuse_Yes 0.02
numpla_3 0.02
othermed_Yes 0.02
numpla_4 0.02
cnctadult_yes 0.02
manrem_Court ordered 0.02
housing_Yes 0.01
numpla_5 0.01
childis_Yes 0.01
curplset_Runaway 0.01
prtsjail_Yes 0.01
curplset_Trial home visit 0.01
dachild_Yes 0.01
relinqsh_Yes 0.01
aaparent_Yes 0.01
vishear_Yes 0.01
mr_Yes 0.01
amiakn_x_yes 0.01
placeout_Yes 0.01
curplset_Supervised independent living 0.01
curplset_Pre-adoptive home 0.01
prtsdied_Yes 0.00
asian_x_yes 0.00
aachild_Yes 0.00
phydis_Yes 0.00
curplset_Missing 0.00
hawaiipi_x_yes 0.00
Feature
0.00
0.02
0.04
0.06
0.08
Feature Importance